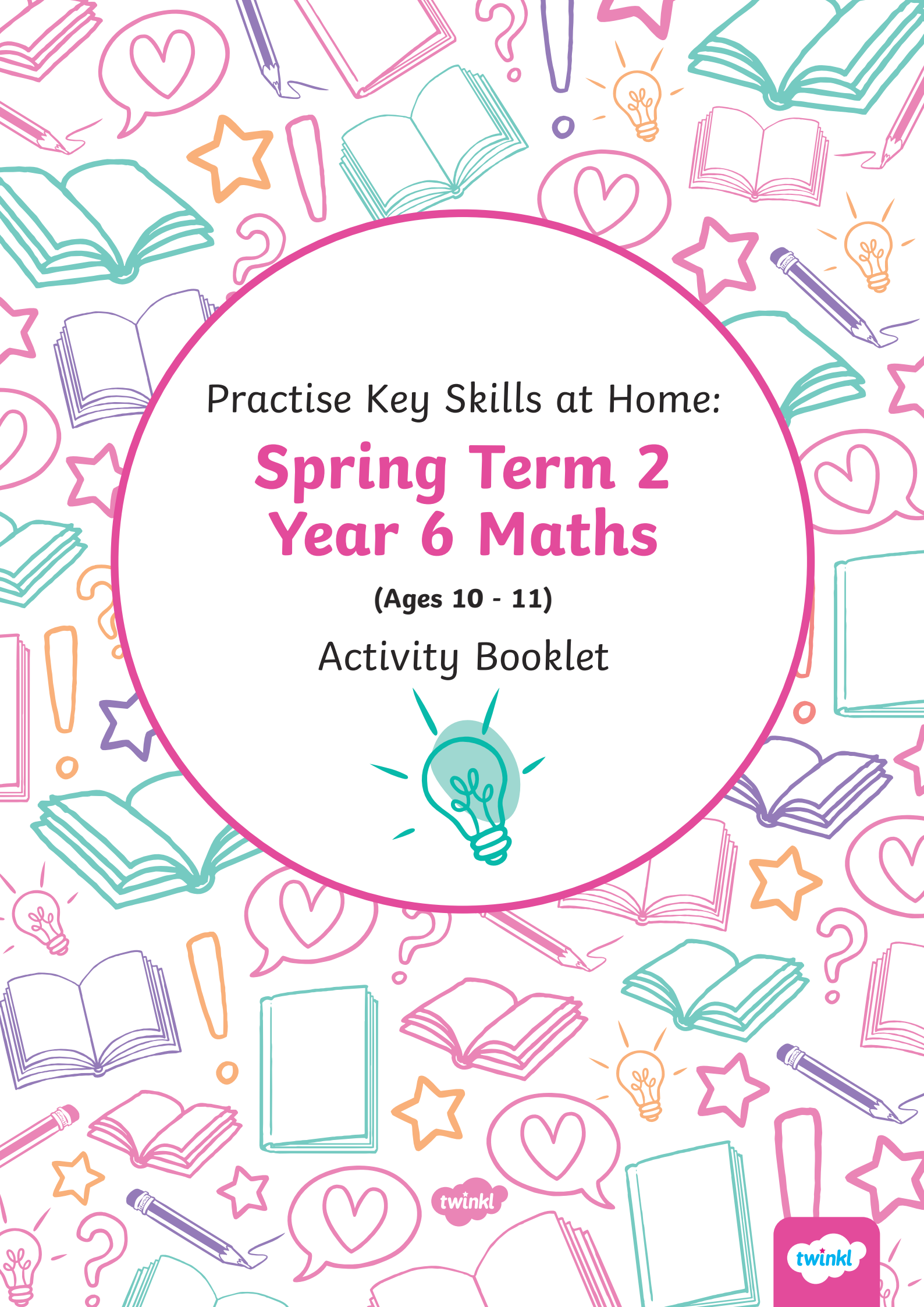




Practise Key Skills at Home:

Spring Term 2 Year 6 Maths

(Ages 10 - 11)

Activity Booklet

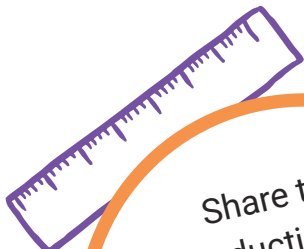


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For Parents

This year 6 activity booklet has been created to help your child practise some of the key skills that they have learnt in primary school. The national curriculum subjects of English and maths have been split among the different booklets. These booklets include clear explanations, examples and activities for your child to try at home in order to help them develop their confidence and understanding of each area.



Share the introduction and explanation for each topic with your child. Talk through it together.

Challenge your child to try the activities independently.



Use the answer pages to encourage your child to self-assess and mark their own activities.

The topics can be done in any order. Your child doesn't have to work through the booklet in order.



Keep the activities short and snappy - your child doesn't need to spend hours on each topic. They are designed to be quick practice activities.



Remember to praise and encourage your child.



Don't worry if your child is finding an activity too difficult; follow the hints and tips on each page to give them some extra support.

How to Use the Booklet

The title of the topic is at the top of each page. This tells your child what skill the activity is practising.

Statistics: Mean Average

There is a handy space at the top of each topic to place a badge or tick when you have completed it!

The mean is the average of a set of data.

To find the mean or average, add up all of the values to find the total. Divide the total by the number of values that you added together. This will give you the mean.

For example, here is a set of data:

6 5 9 8 8 5 9 6

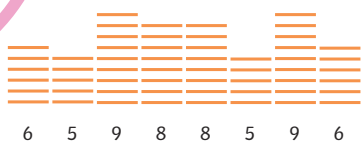
To find the mean, we need to add up all the values.

$$6 + 5 + 9 + 8 + 8 + 5 + 9 + 6 = 56$$

Then, we divide this amount by the number of values in the set of data. There were 8 values in the data set, so $56 \div 8 = 7$

7 means the average of this set of data is 7.

It works because we are 'flattening out' the values. These images show the set of data in the mean average. 8×7 gives the same total as the initial data set. By finding the mean, we've 'evened out' the bumps in the data. It takes into account every value.



The first part of each activity gives your child an explanation of what the topic is about.

The next section gives your child some examples of how to use the skill.

How to Use the Booklet

Finally, there will be an activity for your child to practise applying the skill.

Calculating the Mean

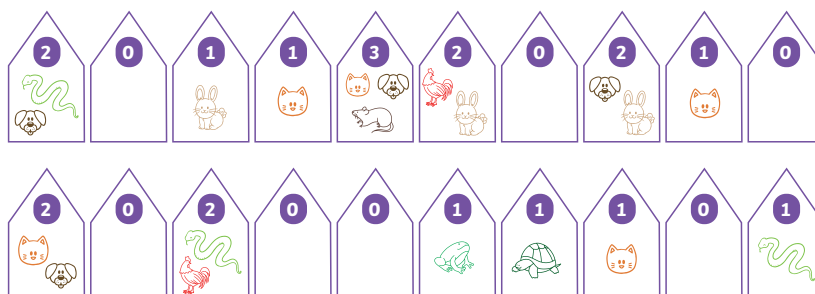
Can you calculate the mean average for each of these data sets?

a. 7 10 6 5 7

b. 14 15 18 5

c. 12 13 2 2 6

Here is a street of houses, showing the number of pets they have. What is the average number of pets per household?



Parent Tip: You can calculate the mean with packets of multi-coloured sweets. Count how many of each colour there are and note this down. Then, add up each value and divide this total by the number of colours to give the average amount of sweets per colour.

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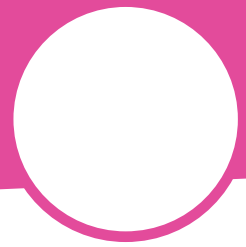
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If your child is finding an activity too difficult, use the parent tip box to give them some extra support.

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Algebra: Simple Formulae



The basic idea behind algebra is to replace a number with a letter or symbol.

For example, we know that $5 \times 2 = 10$.

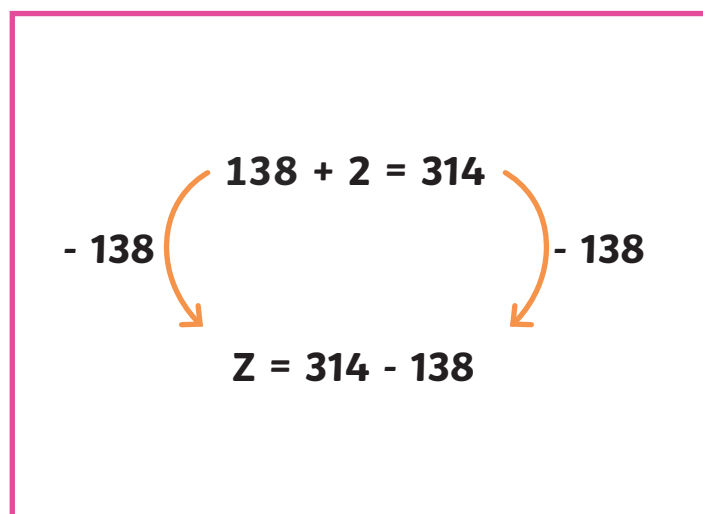
If we were presented with $5 \times ? = 10$, we might know straight away that the missing number is 2. The process we do automatically is $10 \div 5 = 2$, so $? = 2$.

We might be presented with something that appears to be trickier.

For example, $138 + \mathbf{Z} = 314$

Z stands for the unknown number. It could be a **circle** or a **leaf** or a **spiral** or a **cake**... it doesn't matter. It's just there to show that there is an unknown quantity that we need to find.

To solve this, we need to get **z** on its own. To do that, we can see that we could take away 138. We must do the same to both sides of the equation.



You can use any method you like to work out this subtraction. You should be left with...

$$\mathbf{Z} = 176$$

You can check by putting this value back into the equation.

$$138 + 176 = 314$$

$$314 = 314$$

It works; both sides are balanced.

Letters and Numbers

You might have seen something like this:

$$5\mathbf{a} + 2 = 17$$

What does $5\mathbf{a}$ mean?

When we write simple formulae like this, $5\mathbf{a}$ means $5 \times \mathbf{a}$. It indicates that the two amounts are being multiplied together.

So, to solve this one, we would first take away 2 from both sides of the equation...

$$5a + 2 = 17$$

$$5a = 17 - 2$$

$$5a = 15$$

Then, to get a on its own, we need to divide it by 5. We have to do the same for the other side.

$$a = 3$$

You can check it by putting this value back into the equation.

$$(5 \times 3) + 2 = 17$$

$$15 + 2 = 17$$

$$17 = 17$$

Both sides are balanced!

Solving Simple Formulae

Can you solve these algebra problems? Remember, you need to get the unknown on its own. Always do the same to both sides of the equation.

a. $20 = 4a + 4$

b. $3b + 5 = 11$

c. $14 = 6c - 4$

d. $48 = 12d$

e. $2e - 5 = 5$

f. $21 = 3f + 3$

If $g = 6$, can you work out what h is?

$$h = 5g + 14$$

Algebra: Number Sequences and Patterns



Algebra can be very useful when explaining patterns or sequences.

For example, let's look at this sequence of numbers:

1 st term	2 nd term	3 rd term	4 th term
8	11	14	17

We are given the formula $3n + 5$ for this sequence. We would multiply the term value by 3 then add 5. This means that we can use this formula to calculate any number in the sequence.

Without using the formula, we can see that this sequence goes up by 3 each time. Therefore, we can guess that the next number in this sequence (the 5th term) will be 20.

Let's use the formula to check. We are finding the 5th term, so $n = 5$.

$$3 \times 5 + 5$$

$$15 + 5 = 20$$

It works!

This formula is useful when finding a distant term. For example, we might be asked to find the 19th number in this sequence. Instead of working out all the other numbers in between, we can put $n = 19$ straight into our formula.

$$3 \times 19 + 5$$

$$57 + 5 = 62$$

Therefore, the 19th term = 62.

Number Sequences

Use your understanding of number sequences to answer these questions.

a.

The formula $4n + 1$ can be used to calculate the value of the terms in this sequence:

5 9 13 17 21

Fill in the missing information in this table:

Term	Calculation	Value
1 st	$4 \times 1 + 1$	5
6 th		
10 th		
20 th	$4 \times 20 + 1$	41

b.

The formula $3n - 1$ can be used to calculate the value of the terms in this sequence:

2 5 8 11 14

Fill in the missing information in this table:

Term	Calculation	Value
1 st	$3 \times 1 - 1$	2
7 th		
12 th		
18 th	$3 \times 18 - 1$	35

3 7 11 15 19

- C. 11 is the third term in this sequence. Circle the formula that could be used to calculate this term.

$3 \times 4 - 1$

$3 \times 5 - 1$

$3 \times 4 + 1$

2 7 12 17 22

- C. 17 is the fourth term in this sequence. Circle the formula that could be used to calculate this term.

$4 \times 4 + 2$

$4 \times 5 - 3$

$4 \times 5 - 2$

Algebra: Missing Number Problems



Algebraic equations can be used to express missing number problems.

Age in an Equation

We could use one to show the difference in age between two siblings. For example, Ben and Eli were born on the same day but 2 years apart. Ben is 2 years older. We could write this as an equation:

$$e + 2 = b$$

If we add 2 to Eli's age, we can work out Ben's age.

When Ben is 31, how old will Eli be?

$$e + 2 = b$$

$$e + 2 = 31$$

$$e = 31 - 2$$

$$e = 29$$

Eli will be 29.

When Eli is 17, how old will Ben be?

$$e + 2 = b$$

$$17 + 2 = b$$

$$19 = b$$

Ben will be 19.

Decorator Costs Equation

Equations can be useful for calculating costs. For example, if a decorator charges £12 per hour, plus £50 for materials, how could we represent this as an equation?

Let's use **h** for hours.

If the decorator charges £12 per hour, we can write $12h$ to show that the number of hours must be multiplied by £12.

The decorator adds on £50 for materials. We can show this by writing + 50 in the equation.

$$12h + 50$$

If the decorator worked 11 hours on a project, what would the cost be? Let's use the equation...

$$12h + 50$$

$$12 \times 11 + 50$$

$$132 + 50$$

£182 total cost

Missing Number Equations

1.a.

A plumber charges £14 per hour and adds on £75 for materials. Can you write an equation to show how to calculate the total cost? Use **h** for hours.

1.b.

How much would you pay the plumber for 15 hours of work?

2.a.

A hairdresser charges £12 per hour but is offering a £5 discount. Write an equation to show this, using **h** for hours.

2.b.

How much would you pay for a 2-hour session?

3.a.

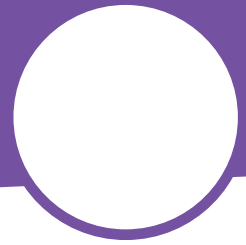
Floor tiles are £24 per square metre. You also need to pay £40 for someone to come and lay the tiles. Write this as an equation, using **m** for square metre.

3.b.

Bhaves'h's kitchen is 5 square metres. How much will he pay for a new tiled floor?

Parent Tip: It helps to highlight the key information in each question, then test out the formula to check if it makes sense.

Algebra: Two Unknowns



Sometimes, there might be two unknowns in an equation. It could be that you have to find all the possible quantities that could fit into the equation.

For example, $ab = 18$.

This means $a \times b = 18$.

We only know that a and b are both whole numbers.

What could these values be?

In this situation, we need to think of pairs of numbers that, when multiplied together, make 18.

Working systematically, we could start with:

$$1 \times 18$$

$$2 \times 9$$

$$3 \times 6$$

After this, we would be swapping the values of a and b around.

$$6 \times 3$$

$$9 \times 2$$

$$18 \times 1$$

These are all the possibilities in a and b are whole numbers.

Sometimes, you might be given a few pieces of information about the unknown values. a and b are, once again, whole numbers.

$$a < 5 \quad 2a + b = 10 \quad b < a$$

From this, we know that a could be 1, 2, 3 or 4.

We also know that b is less than a .

Therefore, a cannot be 1 because there is no whole number less than 1.

Let's test a possible value. If a was 2...

$$2 \times 2 + \mathbf{b} = 10$$

$$4 + \mathbf{b} = 10$$

$$\mathbf{b} = 10 - 4$$

$$\mathbf{b} = 6$$

We know that this can't be right because $\mathbf{b} < \mathbf{a}$ and 6 is not less than 2.

If $\mathbf{a}$ was 3...

$$2 \times 3 + \mathbf{b} = 10$$

$$6 + \mathbf{b} = 10$$

$$\mathbf{b} = 10 - 6$$

$$\mathbf{b} = 4$$

We know that this can't be right because $\mathbf{b} < \mathbf{a}$ and 4 is not less than 3.

If $\mathbf{a}$ was 4...

$$2 \times 4 + \mathbf{b} = 10$$

$$8 + \mathbf{b} = 10$$

$$\mathbf{b} = 10 - 8$$

$$\mathbf{b} = 2$$

This works! 2 is less than 4.

Two Unknowns

a.

Find three possible pairs of numbers for $\mathbf{a}$ and $\mathbf{b}$ in this equation ($\mathbf{a}$ and $\mathbf{b}$ are whole numbers):

$$19 = \mathbf{ab} + 7$$

Value of $\mathbf{a}$	Value of $\mathbf{b}$

b.

Find three possible pairs of numbers for **c** and **d** in this equation (**c** and **d** are whole numbers):

$$13 = \mathbf{cd} - 7$$

Value of **c**

Value of **d**

c.

Find the value of **e** and **f**.

$$\mathbf{ef} = 21 \quad \mathbf{e} + \mathbf{f} = 10 \quad \mathbf{e} < \mathbf{f}$$

e =

f =

d.

Find the value of **g** and **h**.

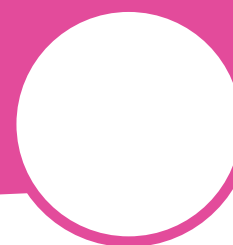
$$\mathbf{g} - \mathbf{h} = 3 \quad \mathbf{g} + \mathbf{h} = 9$$

g =

h =

Parent Tip: It can help to have some scrap paper or a mini whiteboard on hand for scribbling down ideas and testing values in the equation.

Statistics: Tables



Tables are used all the time to collect data (such as a tally chart) or to display data (like a timetable or a score table).

Here is an example of a table showing the favourite drink flavours of children in KS1 and KS2.

Flavour	KS1	KS2	Total
Orange		15	30
Blackcurrant	12	6	
Apple	17	5	
Pineapple	6		
Strawberry		9	
	54	46	100

We can use the information that is already in the table to complete any missing information. For example, we know that there are 46 girls in total. Therefore, we can work out that $15 + 6 + 5 + 9 = 35$, so the missing value for girls who chose pineapple is $46 - 35 = 11$

Here is the completed table:

Flavour	KS1	KS2	Total
Orange	15	15	30
Blackcurrant	12	6	18
Apple	17	5	22
Pineapple	6	11	17
Strawberry	4	9	13
	54	46	100

Let's use this information to answer some questions. This data can tell us about people's preferences.

Which was key stage 1's most popular drink flavour?

We can see that apple flavour was most popular.

Which was key stage 2's most popular drink flavour?

We can see that orange flavour was most popular.

Which was key stage 1's least popular flavour?

The flavour with the fewest votes from the key stage 1 children was strawberry.

Which was key stage 2's least popular flavour?

The flavour with the fewest votes from key stage 2 children was apple.

How many more key stage 1 children chose apple than the key stage 2 children?

17 key stage 1 children chose apple and 5 key stage 2 children chose apple, so $17 - 5 = 12$ more key stage 1 children chose apple than key stage 2 children.

What percentage of children prefer orange?

30 out of 100 children chose orange, so this is 30%.

What fraction of children chose blackcurrant or apple?

18 children chose blackcurrant and 22 children chose apple, so $18 + 22 = 40$. This means $\frac{40}{100}$ or $\frac{2}{5}$ of children chose blackcurrant or apple.

Analysing Data in a Table

Children in key stage one and key stage two were asked to choose their favourite ice cream flavour. Here are the results:

Flavour	KS1	KS2	Total
vanilla	9	11	
chocolate		8	
strawberry	1	10	30
mint chocolate chip	2	13	
cookie dough			24
	38	62	100

1.

First, fill in the missing information.

Next, answer these questions:

- What was key stage 1 children's most popular flavour?
- What was key stage 1 children's least popular flavour?
- What was key stage 2 children's most popular flavour?
- What was key stage 2 children's least popular flavour?
- How many more key stage 1 children chose chocolate flavour than key stage 2 children?
- What percentage of children chose mint chocolate chip?
- What fraction of children chose cookie dough?
- What was the most popular flavour overall?

Parent Tip: Keep an eye out for tables in everyday life, like bus timetables, sports scoreboards or conversion charts. These are all great for practising reading information in table form.

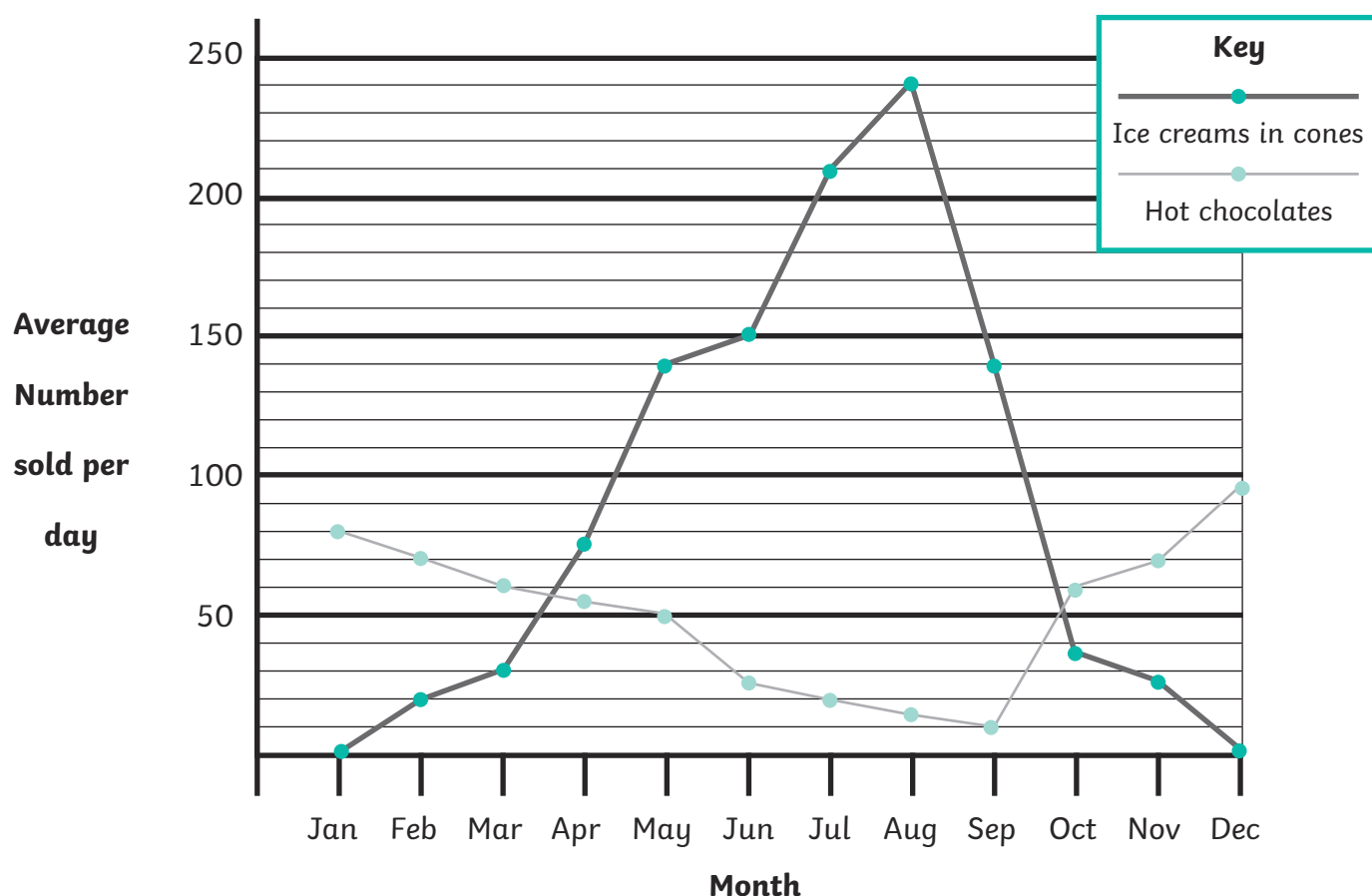
Statistics: Line Graphs

Line graphs are used to show changes to a measurement over time. Sets of points are joined together to make the line.

Data shown in a line graph is continuous. This means it can be measured on a scale that goes on forever. A good example is age. We can measure a person's age in years, but also months, weeks, days, hours, seconds, milliseconds... and so on.

Here is an example of a line graph. This one shows two sets of data. It shows how many ice creams in cones and how many hot chocolates the ice cream van sells on average per day, during each month of the year.

You can see that the points were plotted for each month. Each point shows the average number of sales per day and the month. The month is along the x-axis and the sales are along the y-axis.



Which was the most popular item sold in August?

We can see that the most popular item sold in August was ice creams in cones. August is the peak of ice cream sales. The average number of hot chocolates sold is very low in August.

How many hot chocolates were sold on average per day in April?

By reading the point in the graph and using the scale on the y-axis to help us, we can see that approximately 55 hot chocolates were sold per day in April.

What was the difference between the average number of ice creams and hot chocolates sold each day in November?

In November, approximately 70 hot chocolates were sold per day and 25 ice creams were sold per day. $70 - 25 = 45$, so the difference was 45.

During how many months did the ice cream van sell more than 100 ice creams on average per day?

In May, June, July, August and September, the ice cream van sold more than 100 ice creams on average per day. This means for 5 months the ice cream sales were above 100 per day.

Ice cream cones come in packs of 50. How many boxes of ice cream cones were needed per day in June?

In June, we can see that there were, on average, 150 ice creams sold per day. 3 lots of 50 fit into 150 (or $150 \div 50 = 3$) so this means 3 boxes of ice cream cones were needed each day in June.

Analysing Data in a Line Graph

Here is some data showing Tom's height as he gets older.

Here is Amrit's height over time.

Age (years)	Height (cm)
2	80
3	88
4	95
5	100
6	110
7	118
8	130

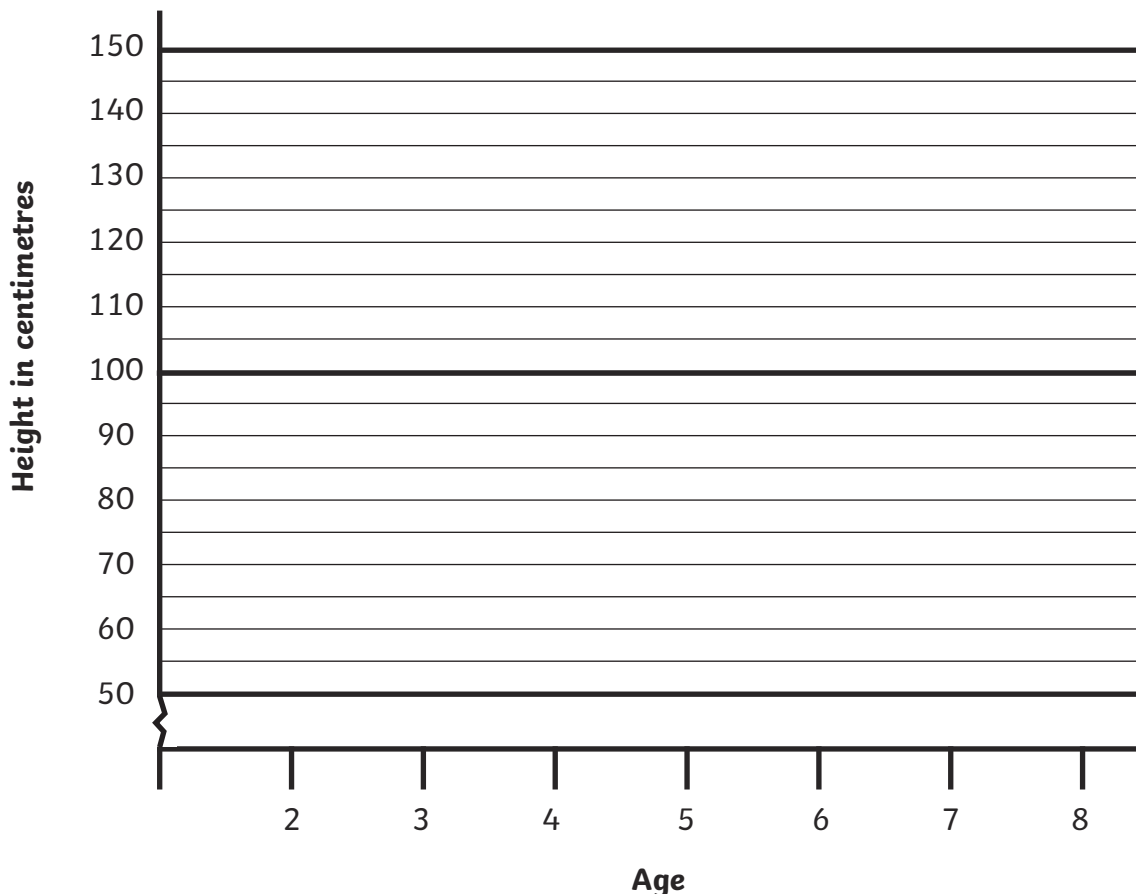
Age (years)	Height (cm)
2	75
3	85
4	100
5	110
6	120
7	125
8	135

Key

Tom =

Amrit =

Plot both sets of data on the same graph, using different colours.



How much taller is 8-year-old Tom compared to 2-year-old Tom?

At approximately what age did Amrit become taller than Tom?

What was the difference between Amrit's height and Tom's height when they were both 6 years old?

What age was Amrit 100cm tall?

What age was Tom 100cm tall?

Parent Tip: Why not record your child's height? This is a fantastic way to practise recording data in a table, then plotting it on a line graph. You might have already been noting down their height each year - even better!

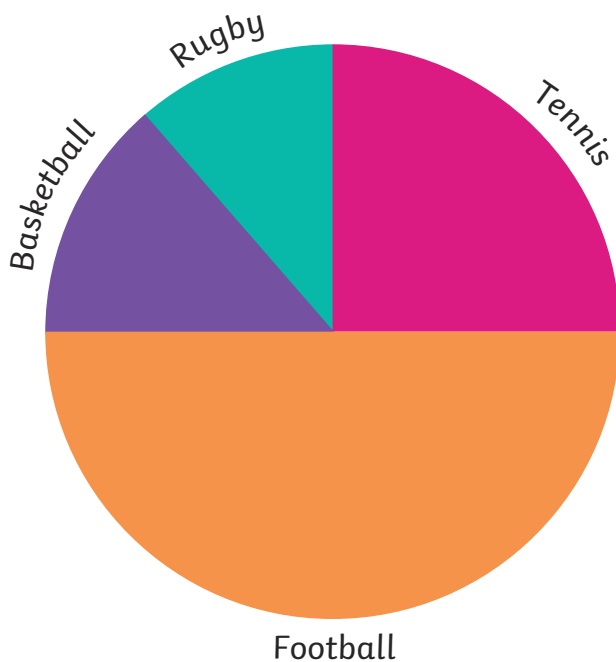
Statistics: Pie Charts



Pie charts represent discrete data. Discrete data means set values. For example, the number of children in a class.

A circle is divided into segments, where each segment represents a data category. The size of each segment matches its proportion of the total amount. Drawing a pie chart requires an understanding of angles, fractions and percentages.

Here is a pie chart that shows 24 people's favourite sports.



The whole circle shows all 24 people. Each segment of the pie represents a sport. We can very quickly see the most popular sport - football - because it takes up the biggest section of the pie chart.

What percentage of people said that basketball was their favourite sport?

We can see that half (50%) chose football and a quarter (25%) chose tennis. The rugby and basketball segments are the same size, so they must be half of a quarter - an eighth - of the total number of people. This is 12.5%.

How many people said that football was their favourite sport?

We know that there are 24 people represented in this pie chart. Half of the pie chart shows people who chose football. Therefore, if we halve 24, we should get the number of people who chose football. This is 12 people.

How many more people chose tennis as their favourite sport than rugby?

We know that 25% or a quarter of people chose tennis. $\frac{1}{4}$ of 24 is $24 \div 4 = 6$.

We know that $\frac{1}{8}$ of people chose rugby. $\frac{1}{8}$ of 24 is $24 \div 8 = 3$.

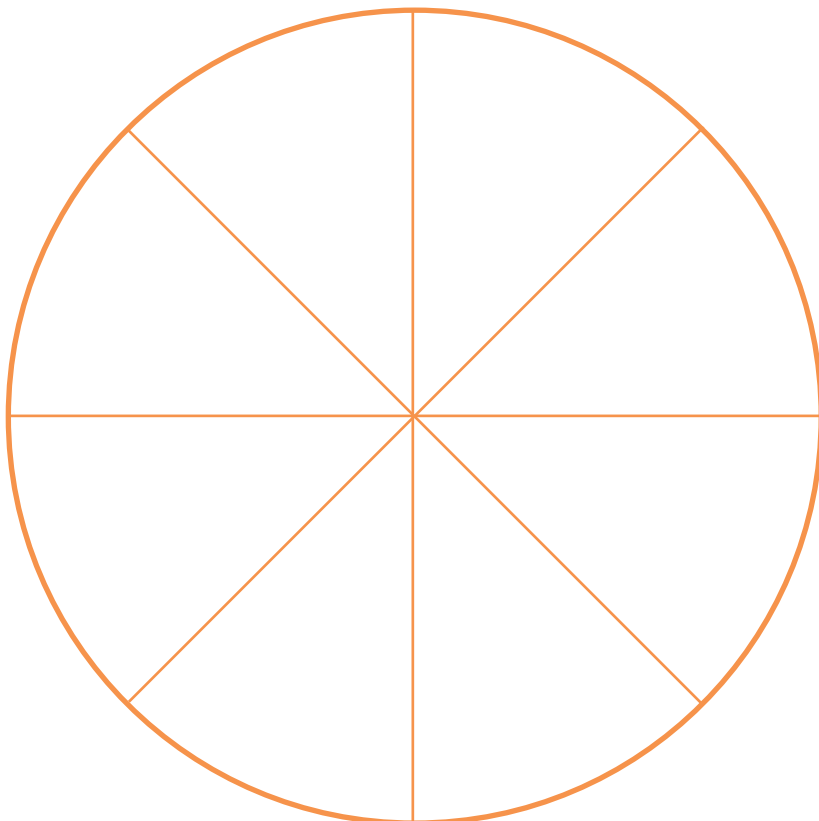
The difference between 6 and 3 is 3, so 3 more people chose tennis over rugby.

Pie Charts

24 children were asked to name their favourite fruit.

Type of fruit	Number of children
Banana	6
Apple	3
Peach	12
Mango	3

Record this data in the pie chart. Make sure you colour in the key to show which colour you are using for each fruit.



Key

Banana	
Apple	
Peach	
Mango	

What percentage of children said peach was their favourite fruit?

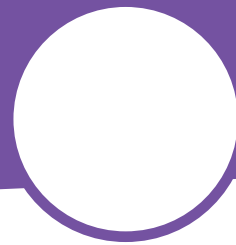
What was the difference between the number of children who said banana compared to mango?

What fraction of children said banana was their favourite fruit?

What percentage of children said apple was their favourite fruit?

Parent Tip: Recreate a pie chart using a pizza, a cake or a pie. You could use different toppings or decorations to show the different segments.

Statistics: Mean Average



The mean is the average of a set of data.

To find the mean or average, add up all of the values to find the total. Divide the total by the number of values that you added together. This will give you the mean.

For example, here is a set of data:

6 5 9 8 8 5 9 6

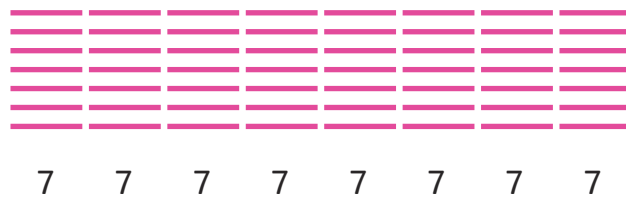
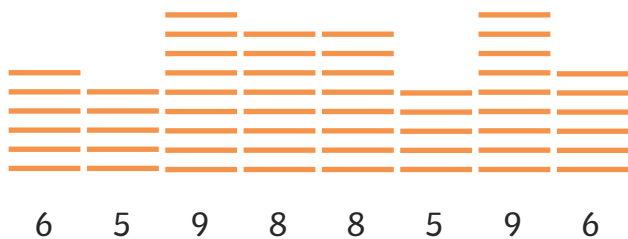
To find the mean, we need to add up all the values.

$$6 + 5 + 9 + 8 + 8 + 5 + 9 + 6 = 56$$

Then, we divide this amount by the number of values in the set of data. There were 8 values in the data set, so $56 \div 8 = 7$

This means the average of this set of data is 7.

This works because we are 'flattening out' the values. These images show the set of data and then the mean average. 8×7 gives the same total as the initial data set. By finding the average, we've 'evened out' the bumps in the data. It takes into account every value.



Calculating the Mean

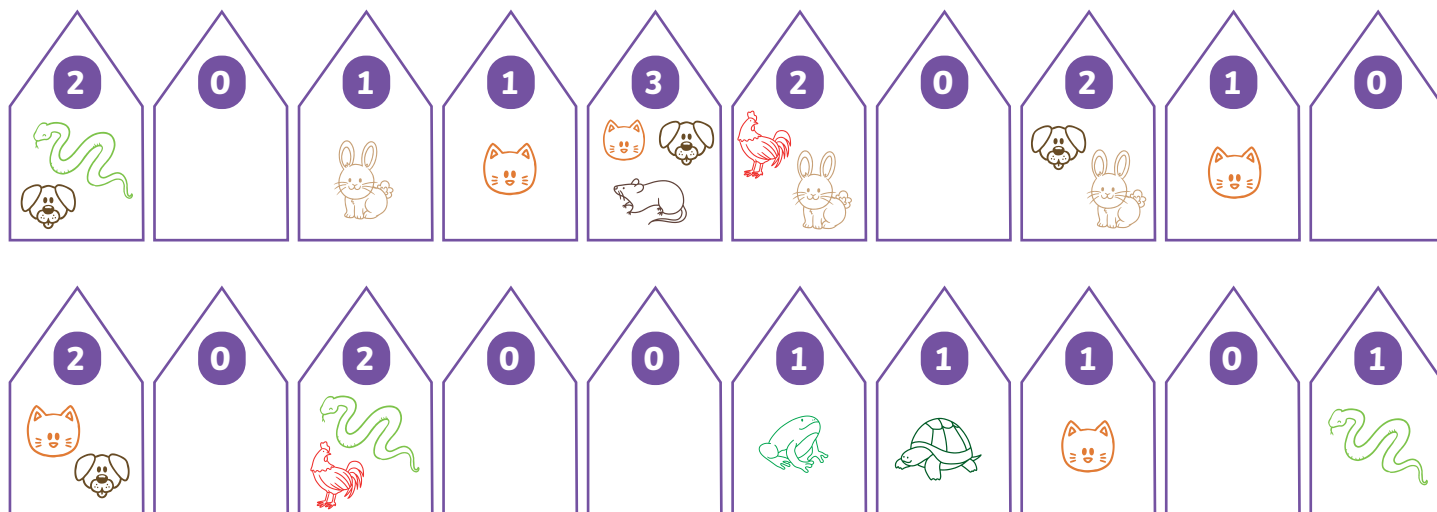
Can you calculate the mean average for each of these data sets?

a. 7 10 6 5 7

b. 14 15 18 5

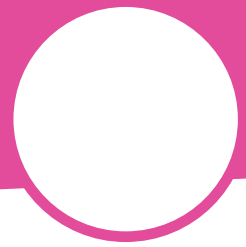
c. 12 13 2 2 6

Here is a street of houses, showing the number of pets they have. What is the average number of pets per household?



Parent Tip: You can calculate the mean with packets of multi-coloured sweets. Count how many of each colour there are and note this down. Then, add up each value and divide this total by the number of colours to give the average amount of sweets per colour.

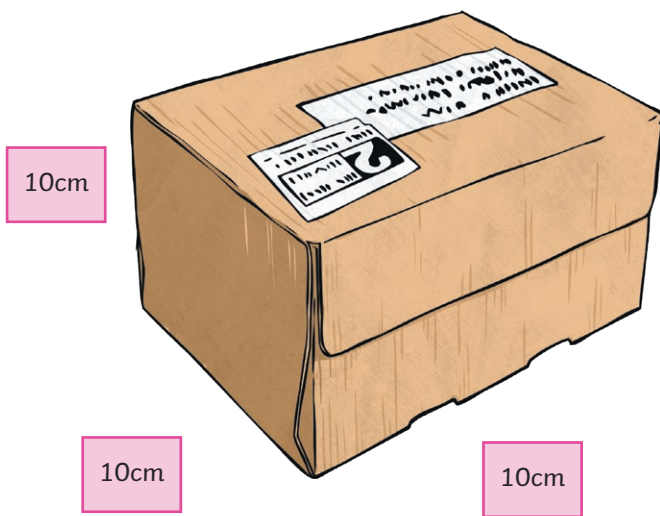
Calculate, Estimate and Compare Area of Cubes and Cuboids



For this key skill, you will need to be able to look at cubes (with their measurements labelled) and estimate which one is largest or smallest. Then calculate the volume to check your answers.

Area is measured in cubic centimetres (cm^3) and cubic metres (m^3).

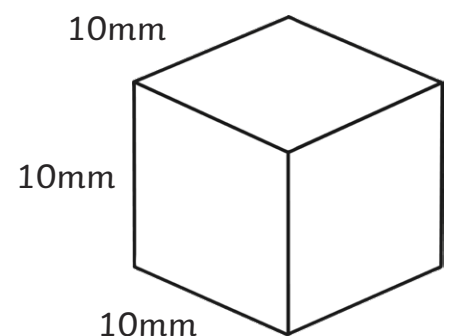
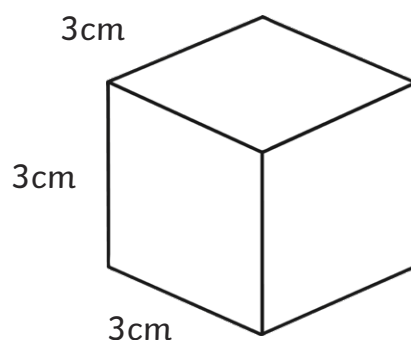
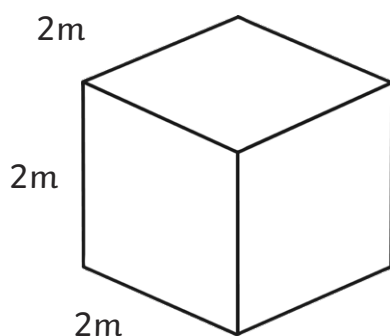
For example, you just need to look at this parcel and know that the formula for volume is length $\times$ height $\times$ width. You could quickly estimate that the volume would be 1000cm^3 .



Calculate, Estimate and Compare Area of Cubes and Cuboids

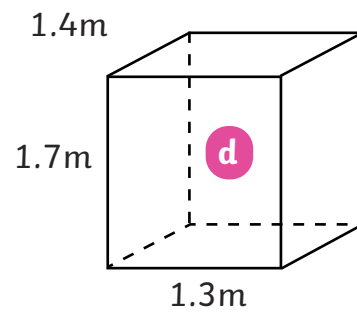
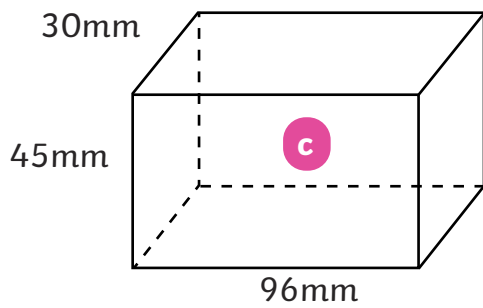
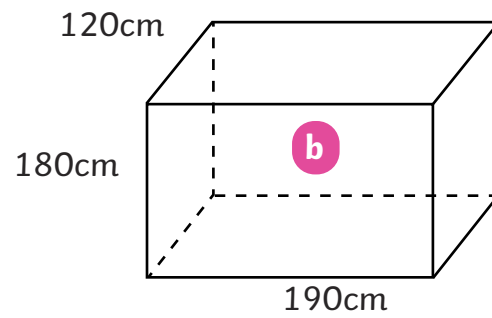
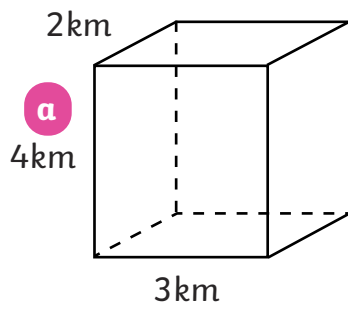
a.

Tick the shape that has the largest volume:



b.

Here are 4 cuboids:



Order the cuboids by volume from largest to smallest

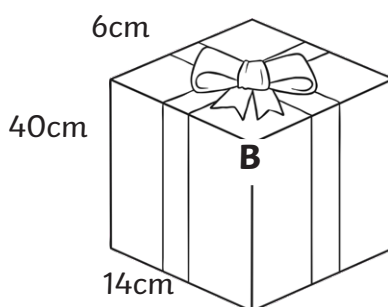
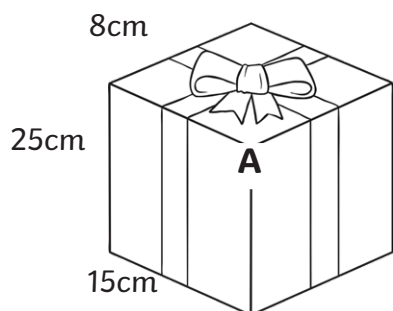
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Largest

Smallest

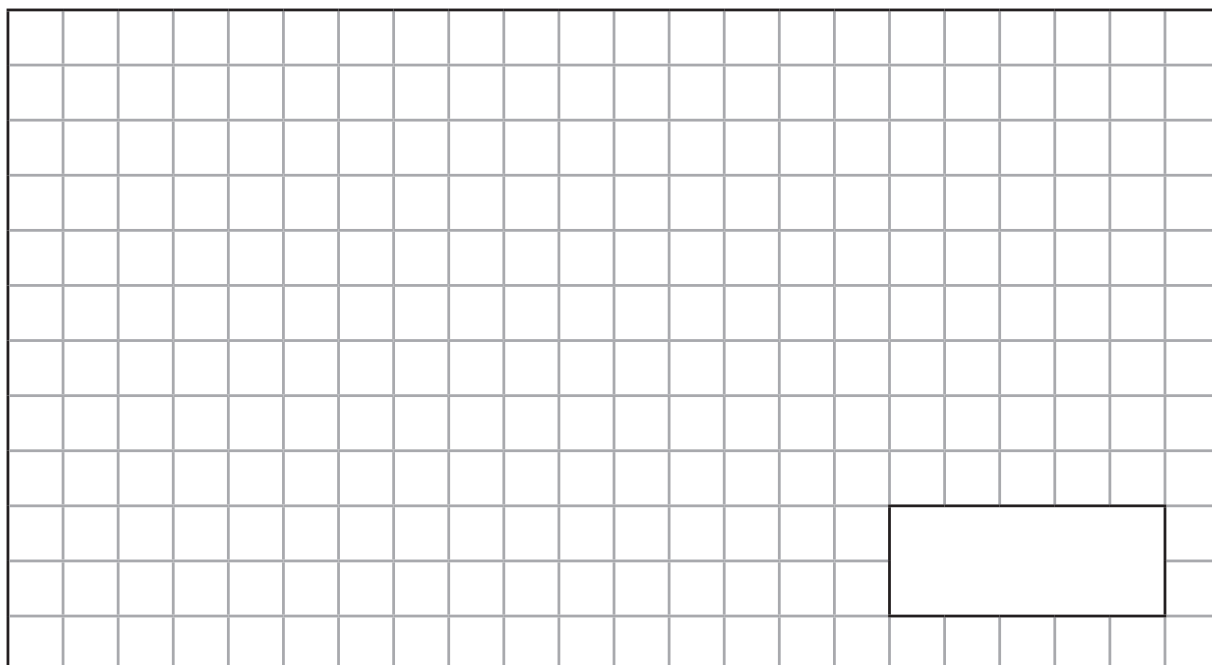
c.

Here are two gift boxes. Shola wants to work out which box has the greater volume.



not drawn to scale

Which gift box has the greater volume?

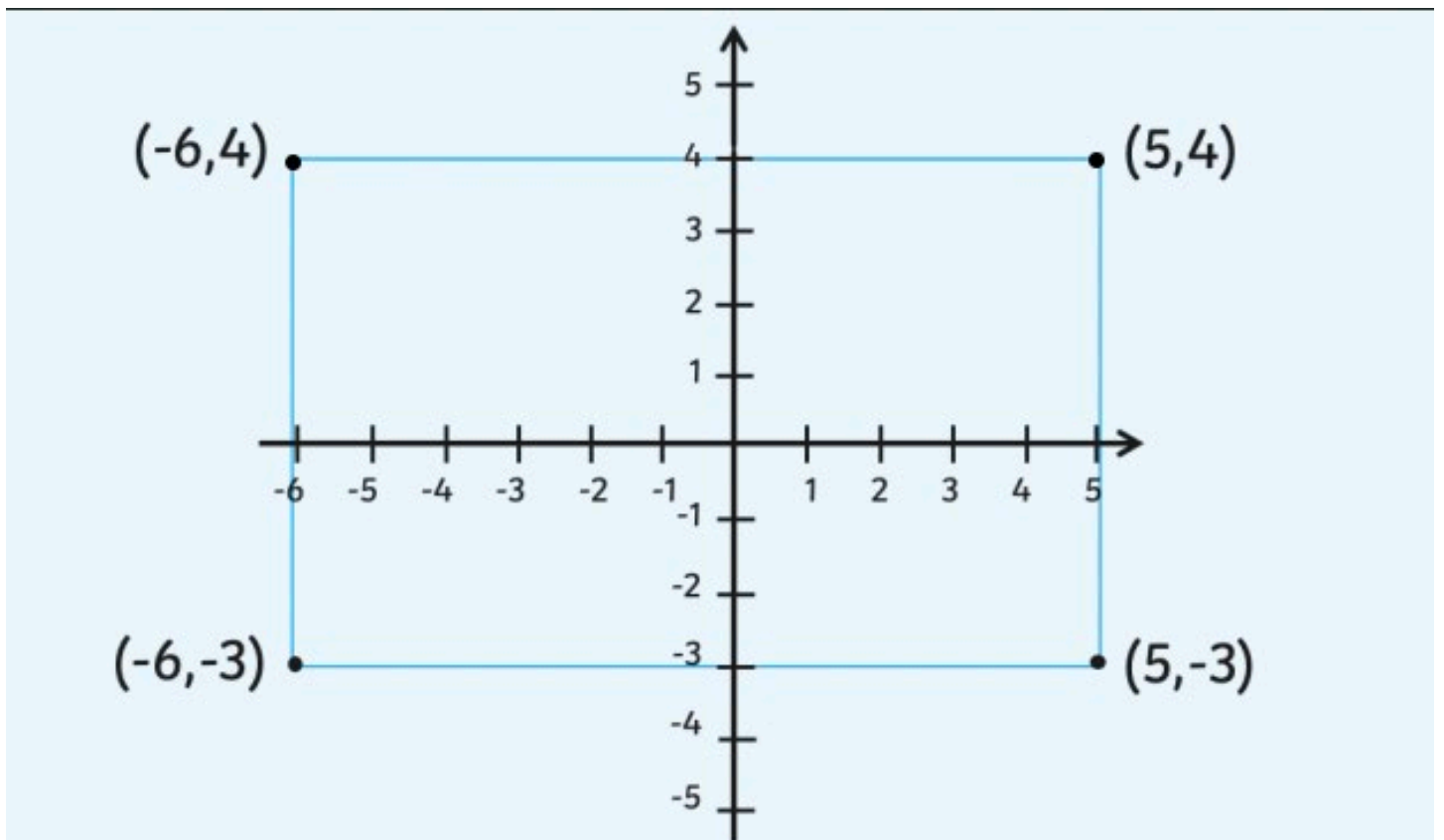


Parent Tip: Although it is not a standard unit of measure, it can be fun to use building blocks to calculate volume. Ask your child to build a box that is 3 bricks high, 4 blocks wide and 3 blocks deep. Can they work out the volume, if they had made a solid block? Check their answers by making the solid block and counting the bricks.

Coordinates

Coordinates allow us to **determine position** using numbers on a grid. By the end of primary school, children will have been introduced to the x and y-axis and will have started working with coordinates in four quadrants. We present coordinates as numbers, inside brackets. For example, **(5, 4)**. The x coordinate always goes first **(x, y)**.

In the first quadrant both x and y are positive. In the second quadrant, x is negative, y is positive. In the third quadrant, x and y are both negative. In the fourth quadrant, x is positive, y is negative.



Children will learn how to use coordinates to describe the position of points and shapes on a grid, as well as plotting points and shapes when given coordinates.

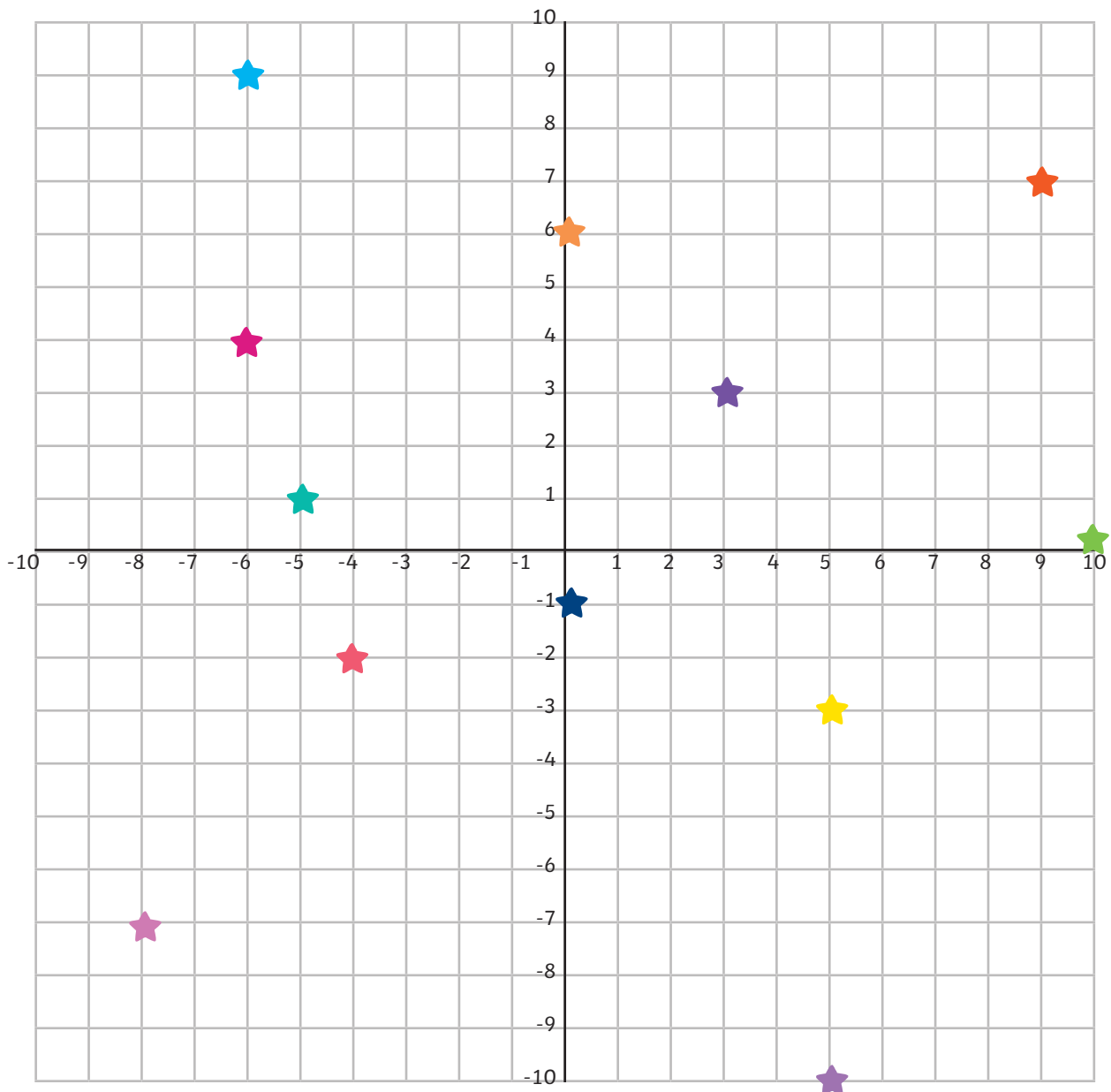
The phrase '**along the corridor, up the stairs**' is often used to help children remember that x comes before y in the coordinates (x, y). This applies when the x and/or the y is negative too (see the diagram above). Some people find it useful to use a ruler along the grid lines to help read the numbers.

Coordinates

a.

Reading Coordinates

Look at the coloured stars on the grid. Write the coordinates. Don't forget (x, y).



b.

Plotting Coordinates

Plot the following sets of coordinates on the grid using a dot or small cross. Connect the dots in each set with straight lines to make a shape.

1. $(-7, -3)$ $(-9, -6)$ $(-2, -6)$ $(-4, -3)$

2. $(-3, 0)$ $(-5, 2)$ $(-7, 0)$ $(-5, -2)$

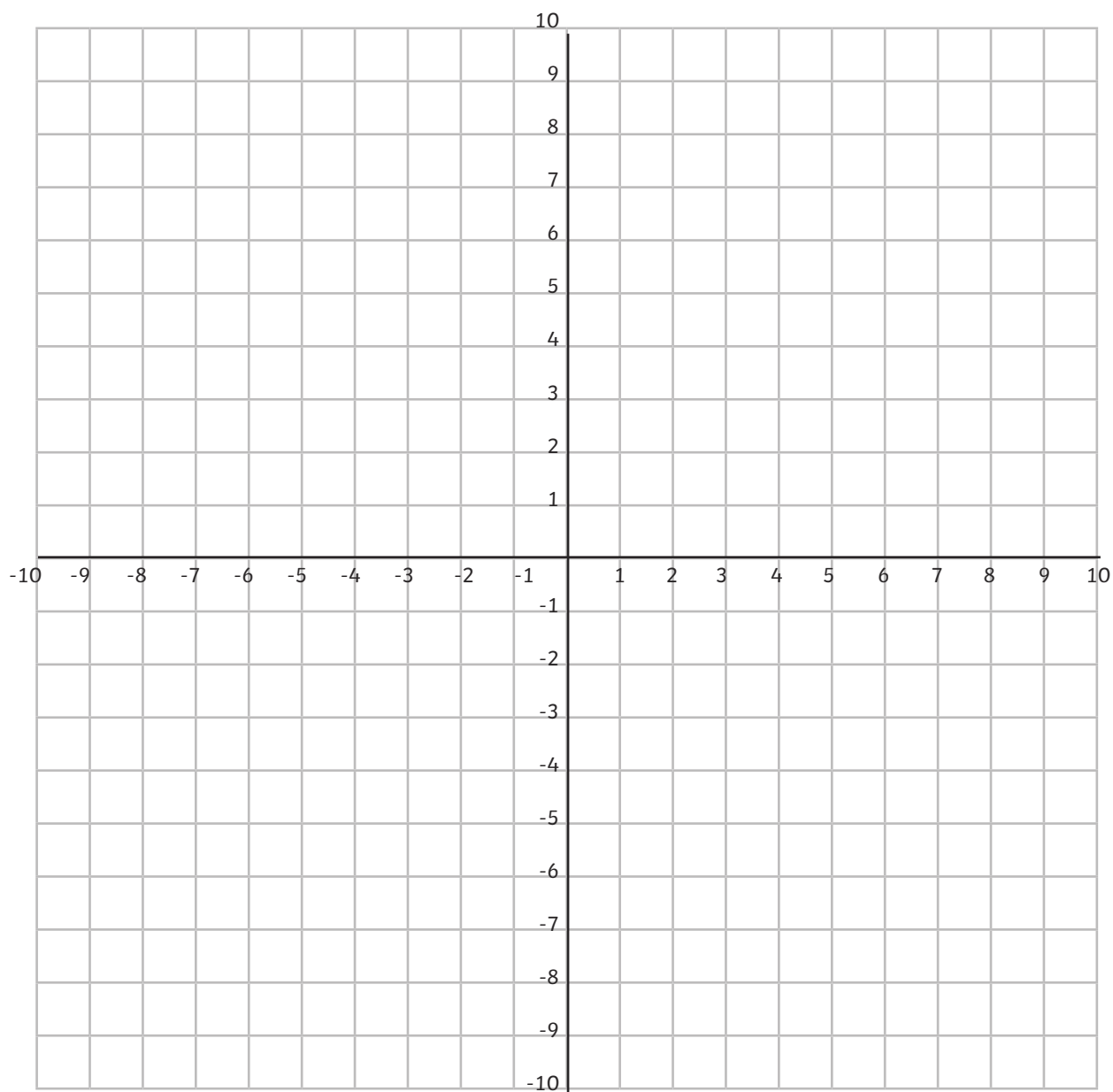
3. $(0, 4)$ $(3, 7)$ $(9, 7)$ $(6, 4)$

4. $(5, -4)$ $(3, -6)$ $(5, -9)$ $(7, -6)$ $(5, -4)$

Name the Quadrilateral

c.

A quadrilateral is any shape with four sides and four angles. This includes squares, rectangles, parallelograms, trapeziums and more. Write the name of each quadrilateral next to the shapes on the grid.

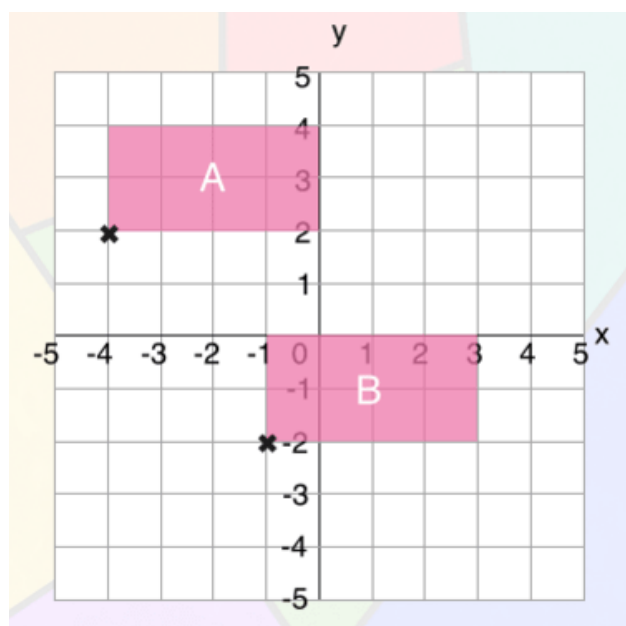


Parent Tip: A ruler is really important for drawing shapes and can be useful for checking coordinates using grid lines too.

Translation and Reflection

When talking about shape, there are different ways in which we can transform a shape.

Translation describes the movement of a shape from one position to another. When a shape is translated, it can move left and right, up and down, but the size and appearance of the shape stay the same. When translating shapes, it can be helpful to choose one point on the shape to concentrate on.

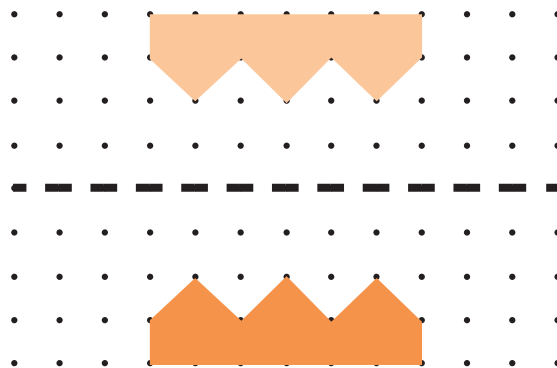
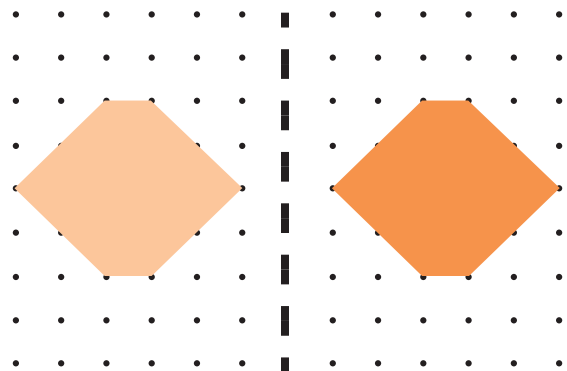
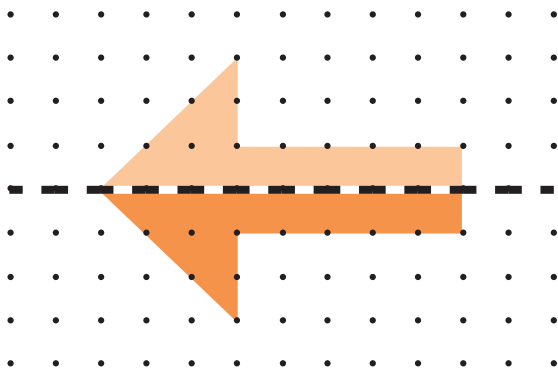


For example, we can look at how the black cross has moved. We would say that the black cross has moved 3 squares to the right and 4 squares down. This is a translation.

Children will learn how to describe the translation of simple 2D shapes, as well as being given a translation to complete themselves. This could be working in one quadrant or four, like the example above. Often children may be asked to give the original and translated coordinates for the shape.

Reflection is when a shape is 'flipped' over a line of reflection to give two mirror images. The shape's appearance and size do not change, but they will look as if the left and right side have been switched. Children will use mirror lines parallel to the x and y-axis and might reflect part of a shape to make a whole shape, reflect a whole shape, complete a reflected pattern or use coordinates to describe and reflect shapes.

Here are a few examples:

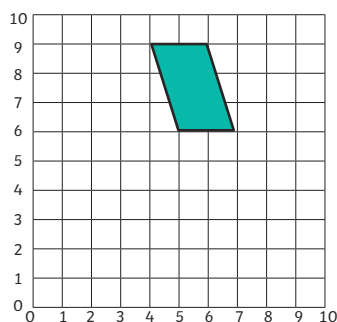


a. Translating Shapes

This parallelogram is translated **left**
3, down 5.

Draw the parallelogram in its
new position.

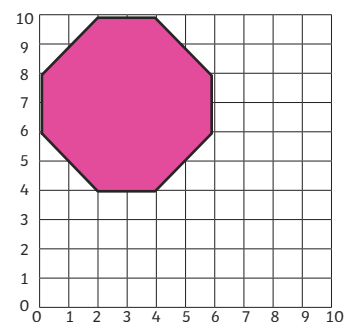
1.



This octagon is translated **right**
4, down 4.

Draw the octagon in its
new position.

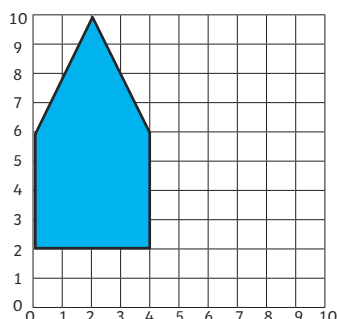
2.



This pentagon is translated **right 6, down 2**.

Draw the pentagon in its new position.

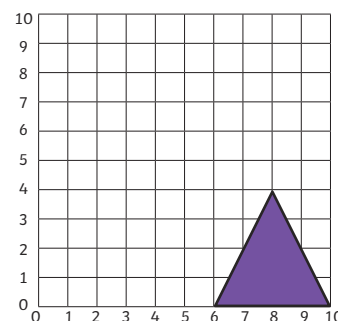
3.



This equilateral triangle is translated **left 5, up 5**.

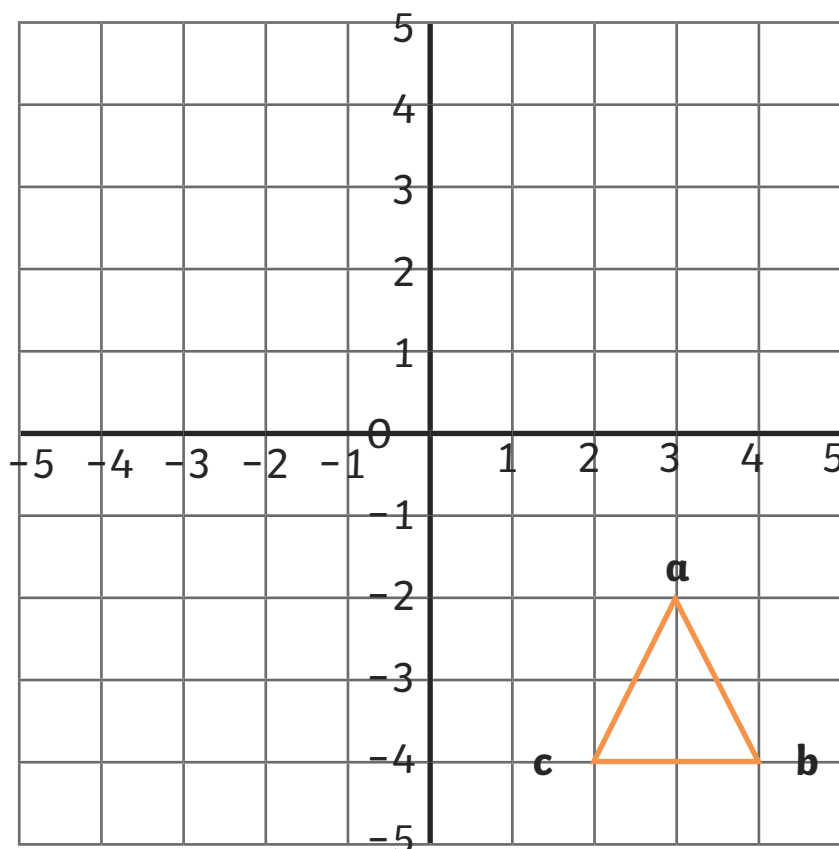
Draw the triangle in its new position.

4.



Parent Tip: Once your child has translated the shapes, can they describe the properties of each shape? They could try naming the shape, think about how many sides it has, identify whether it is regular or irregular, identify the types of lines (vertical, horizontal, parallel, perpendicular) they can see, count the vertices and see if they can spot any lines of symmetry.

b. Translation and Coordinates



Using the triangle, answer the following questions.

If the triangle is translated 5 squares left, what will the new coordinates of vertex A be?

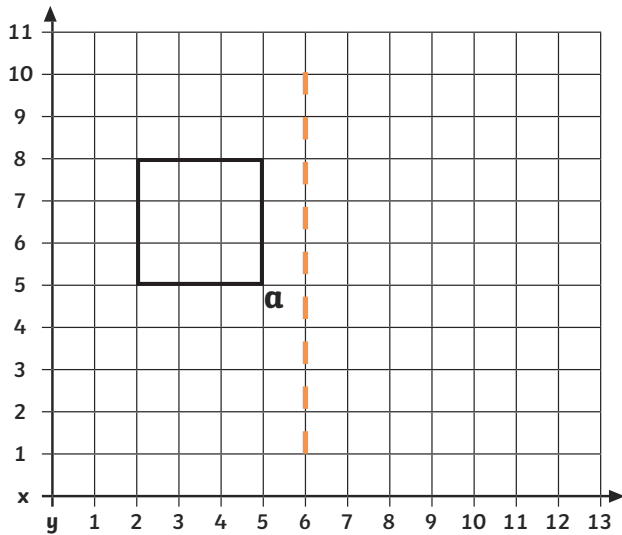
If the triangle is translated 3 squares up, what will the new coordinates of vertex B be?

If the triangle is translated four squares left and 6 squares up, what will the new coordinates be for vertex C?

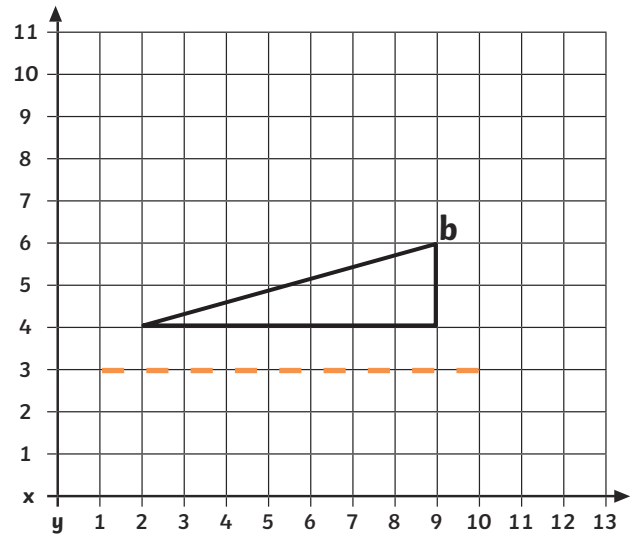
c. Reflecting Shapes

Use the mirror line to reflect the shape and give the new coordinate.

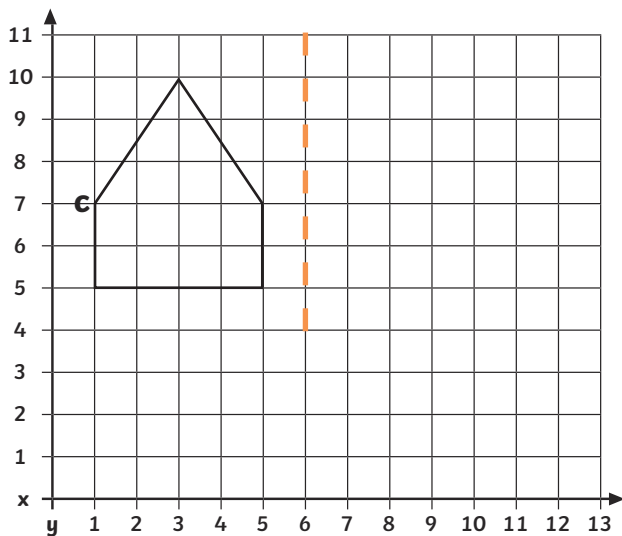
1) New coordinates of corner a = _____



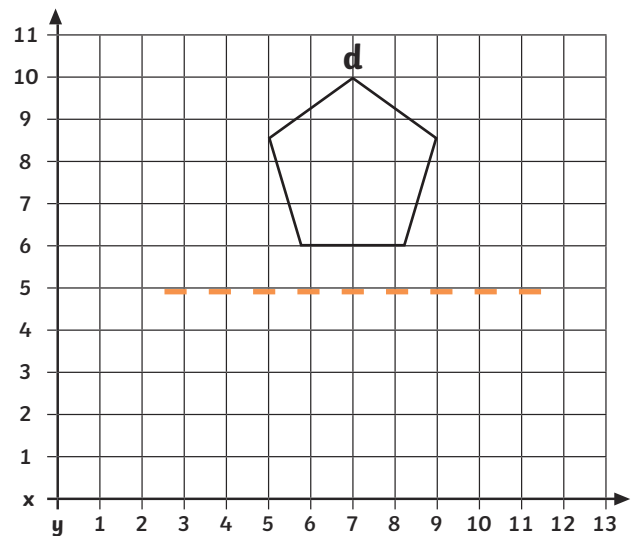
2) New coordinates of corner b = _____

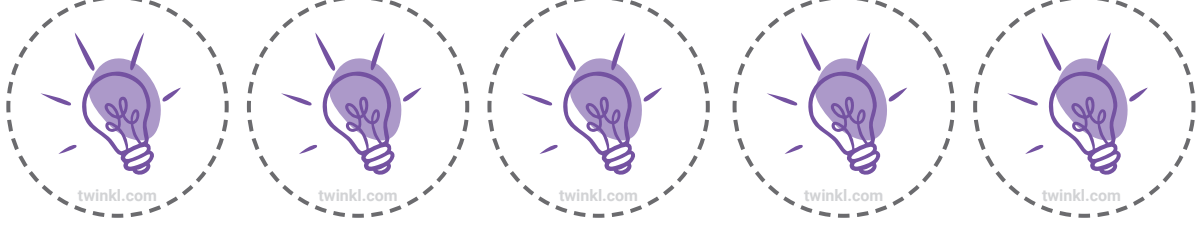
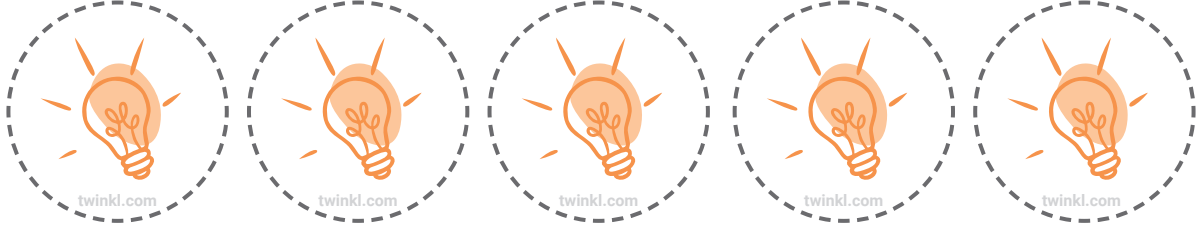
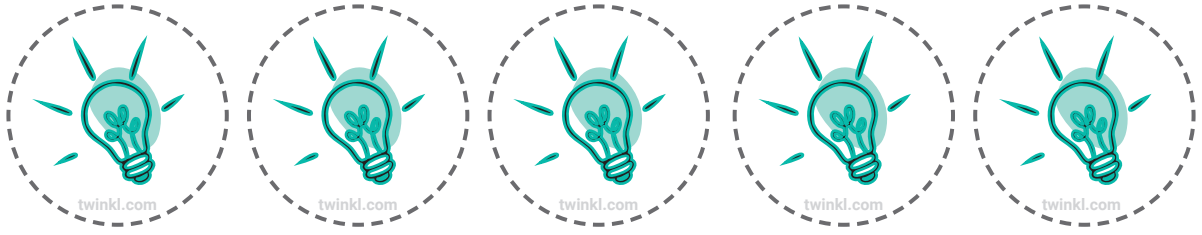


3) New coordinates of corner c = _____



4) New coordinates of corner d = _____





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Imagine resources are designed to help your children to think creatively, question and imagine. Every week, a new topic consisting of five photos, each with related activities, is created.

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Algebra: Solving Simple Formulae - Answers

Can you solve these algebra problems? Remember, you need to get the unknown on its own. Always do the same to both sides of the equation.

$$a = 4$$

$$b = 2$$

$$c = 3$$

$$d = 4$$

$$e = 5$$

$$f = 6$$

If $g = 6$, can you work out what h is?

$$h = 5g + 14$$

$$h = 30 + 14$$

$$\mathbf{h = 44}$$

Algebra: Number Sequences - Answers

a.

The formula $4n + 1$ can be used to calculate the value of the terms in this sequence:

5 9 13 17 21

Fill in the missing information in this table:

Term	Calculation	Value
1 st	$4 \times 1 + 1$	5
6 th	$4 \times 6 + 1$	25
10 th	$4 \times 10 + 1$	41
20 th	$4 \times 20 + 1$	81

b.

The formula $3n - 1$ can be used to calculate the value of the terms in this sequence:

2 5 8 11 14

Fill in the missing information in this table:

Term	Calculation	Value
1 st	$3 \times 1 - 1$	2
7 th	$3 \times 7 - 1$	20
12 th	$3 \times 12 - 1$	35
18 th	$3 \times 18 - 1$	53

3 7 11 15 19

- C.** 11 is the third term in this sequence. Circle the formula that could be used to calculate this term.

$3 \times 4 - 1$

$3 \times 5 - 1$

$3 \times 4 + 1$

2 7 12 17 22

- C.** 17 is the fourth term in this sequence. Circle the formula that could be used to calculate this term.

$4 \times 4 + 2$

$4 \times 5 - 3$

$4 \times 5 - 2$

Algebra: Missing Number Problems - Answers

1.

a. $14h + 75$

b. £285

2.

a. $12h - 5$

b. £19

3.

a. $24m + 40$

b. £160

Algebra: Two Unknowns - Answers

Any three of these possibilities:

a. Find three possible pairs of numbers for **a** and **b** in this equation (**a** and **b** are whole numbers):

$$19 = \mathbf{ab} + 7$$

Value of a	Value of b
1	12
2	6
3	4
4	3
6	2
12	1

Any three of these possibilities:

b. Find three possible pairs of numbers for **c** and **d** in this equation (**c** and **d** are whole numbers):

$$13 = \mathbf{cd} - 7$$

Value of c	Value of d
1	20
2	10
4	5
5	4
10	2
20	1

c. Find the value of **e** and **f**.

$\mathbf{ef} = 21$ $\mathbf{e} + \mathbf{f} = 10$ $\mathbf{e} < \mathbf{f}$	$\mathbf{e} = 3$ $\mathbf{f} = 7$
---	-----------------------------------

d. Find the value of **g** and **h**.

$\mathbf{g} - \mathbf{h} = 3$ $\mathbf{g} + \mathbf{h} = 9$	$\mathbf{g} = 6$ $\mathbf{h} = 3$
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Algebra: Two Unknowns - Answers

Flavour	KS1	KS2	Total
vanilla	9	11	20
chocolate	22	8	30
strawberry	1	10	11
mint chocolate chip	2	13	15
cookie dough	4	20	24
	38	62	100

1.

First, fill in the missing information.

Next, answer these questions:

What was key stage 1 children's most popular flavour? **Chocolate**

What was key stage 1 children's least popular flavour? Strawberry

What was key stage 2 children's most popular flavour? **Cookie dough**

What was key stage 2 children's least popular flavour? **Chocolate**

How many more key stage 1 children chose chocolate flavour than key stage 2 children?

22 - 8 = 14 more

What percentage of children chose mint chocolate chip? **15%**

What fraction of children chose cookie dough? $\frac{24}{100}$ or $\frac{6}{25}$

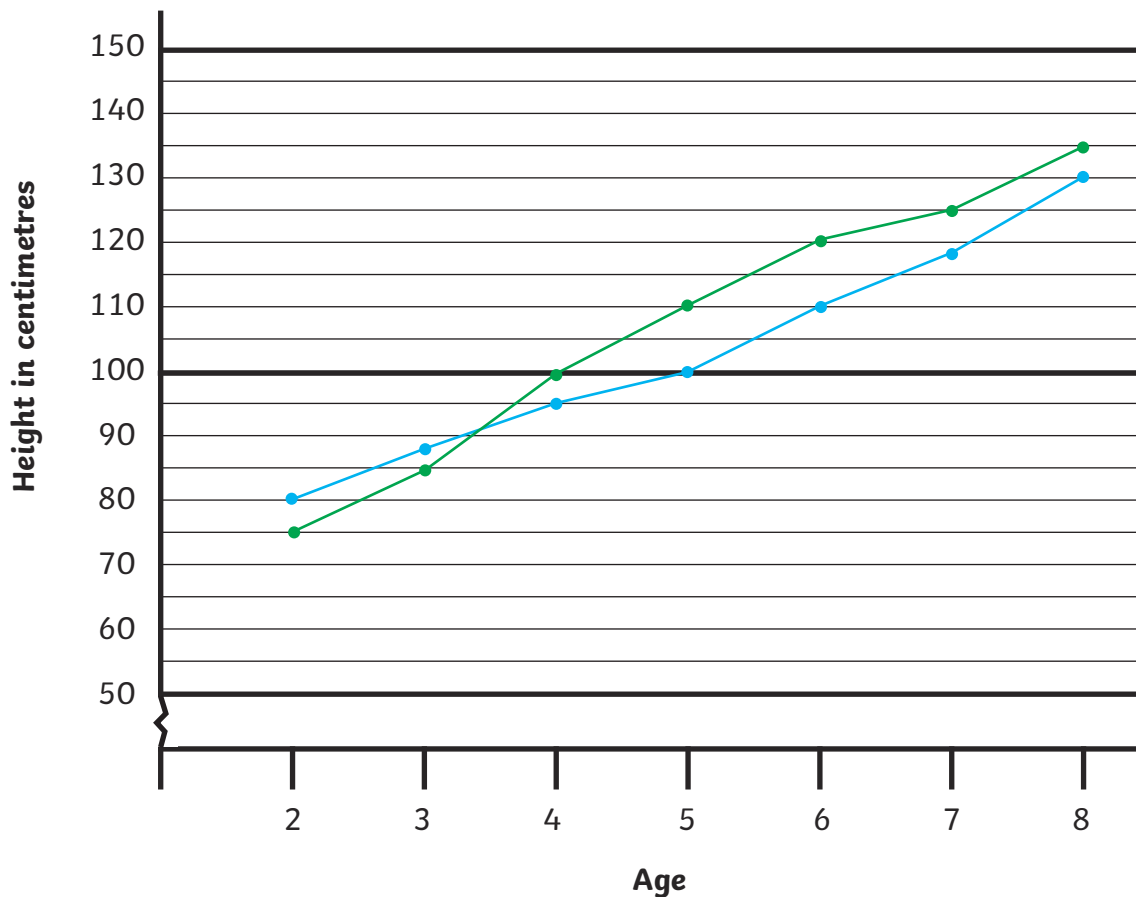
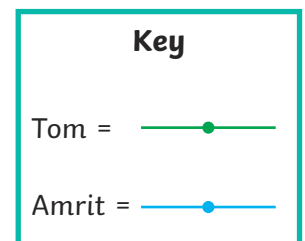
What was the most popular flavour overall? **Chocolate**

Statistics: Analysing Data in a Line Graph - Answers

Age (years)	Height (cm)
2	80
3	88
4	95
5	100
6	110
7	118
8	130

Age (years)	Height (cm)
2	75
3	85
4	100
5	110
6	120
7	125
8	135

Plot both sets of data on the same graph, using different colours.



How much taller is 8-year-old Tom compared to 2-year-old Tom?

130cm - 80cm = 50cm taller

At approximately what age did Amrit become taller than Tom?

4 years old (or between the ages of 3 and 4 years old)

What was the difference between Amrit's height and Tom's height when they were both 6 years old?

120cm - 110cm = 10cm difference

What age was Amrit 100cm tall?

4 years old

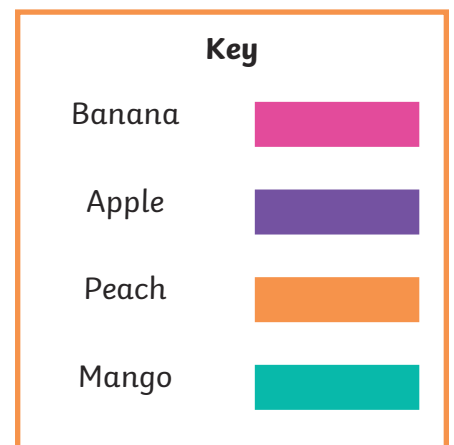
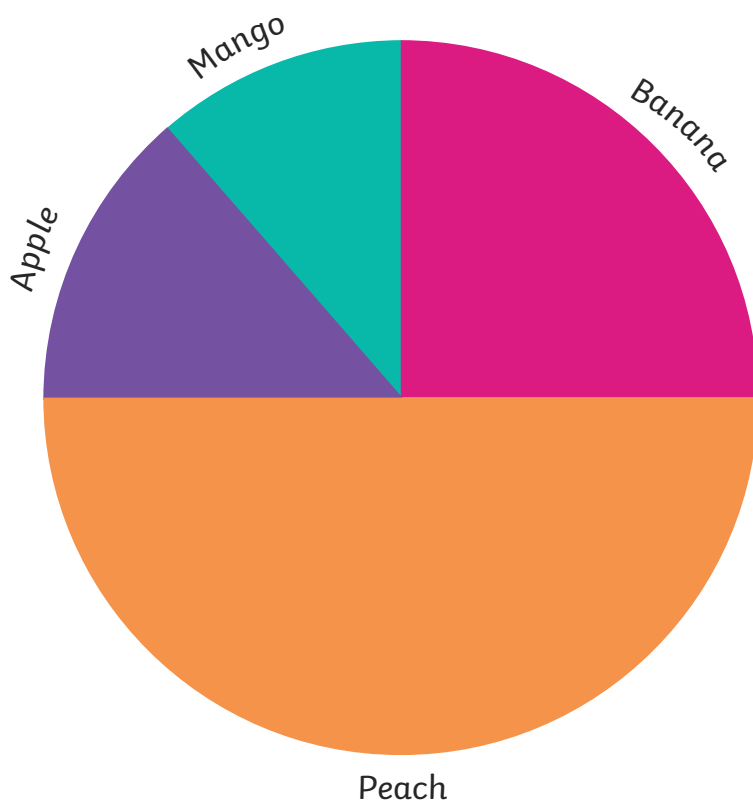
What age was Tom 100cm tall?

5 years old

Statistics: Pie Charts - Answers

Type of fruit	Number of children
Banana	6
Apple	3
Peach	12
Mango	3

Record this data in the pie chart. Make sure you colour in the key to show which colour you are using for each fruit.



What percentage of children said peach was their favourite fruit?

50%

What was the difference between the number of children who said banana compared to mango?

$$6 - 3 = 3$$

What fraction of children said banana was their favourite fruit?

$$\frac{1}{4}$$

What percentage of children said apple was their favourite fruit?

12.5%

Statistics: Calculating the Mean - Answers

a. 7 10 6 5 7

$$\begin{array}{r} 7 \quad 10 \quad 6 \quad 5 \quad 7 \\ = 35 \div 5 = 7 \end{array}$$

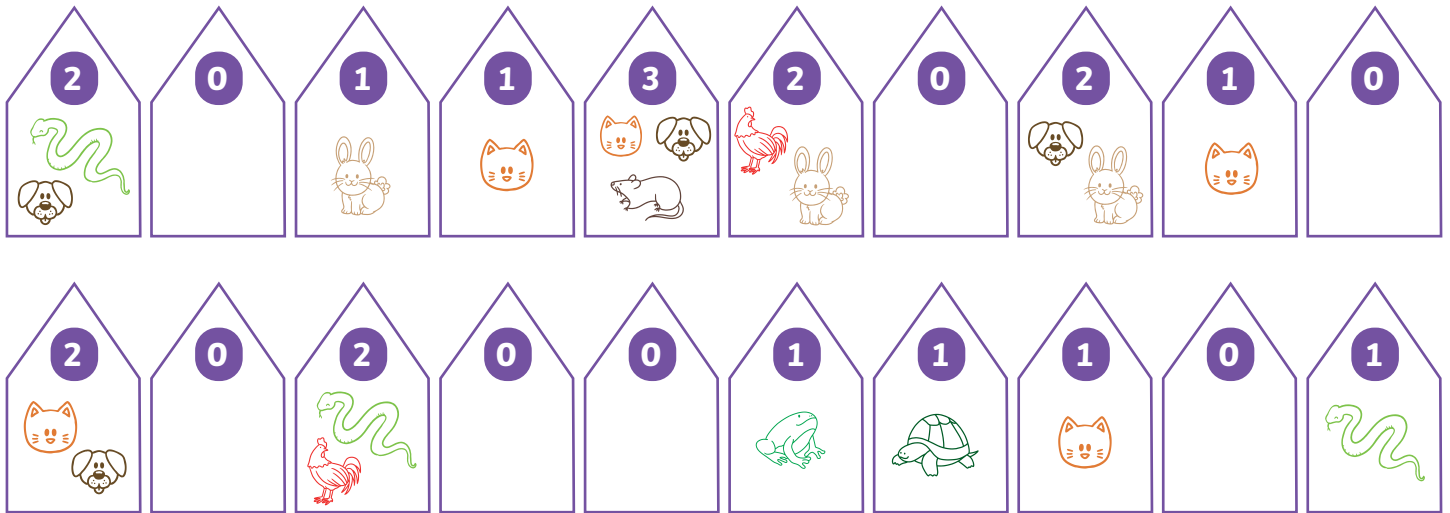
b. 14 15 18 5

$$\begin{array}{r} 14 \quad 15 \quad 18 \quad 5 \\ = 52 \div 4 = 13 \end{array}$$

c. 12 13 2 2 6

$$\begin{array}{r} 12 \quad 13 \quad 2 \quad 2 \quad 6 \\ = 35 \div 5 = 7 \end{array}$$

Here is a street of houses, showing the number of pets they have. What is the average number of pets per household?



$$2 + 0 + 1 + 1 + 3 + 2 + 0 + 2 + 1 + 0 + 2 + 0 + 2 + 0 + 0 + 1 + 1 + 1 + 0 + 1 = 20$$

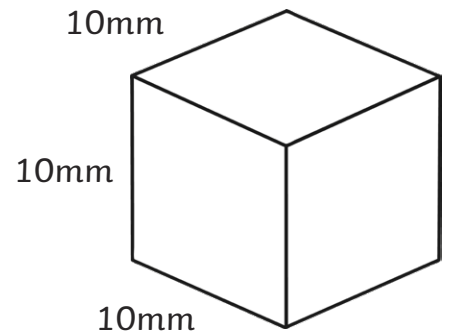
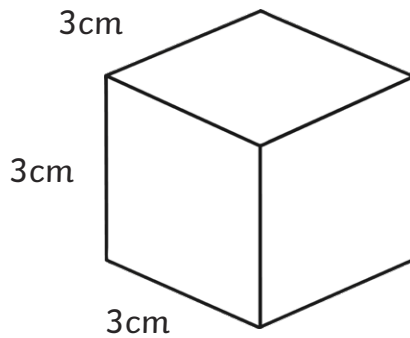
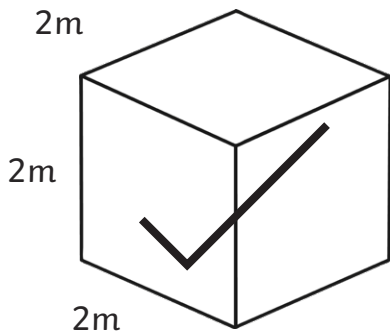
$$20 \div 20 \text{ houses} = 1$$

The mean average is **1 pet per household.**

Calculate, Estimate and Compare Area of Cubes and Cuboids - Answers

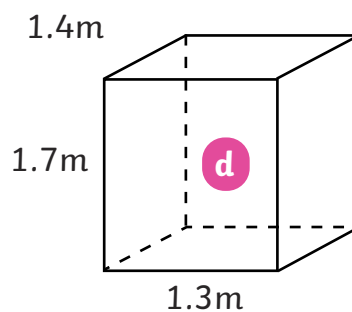
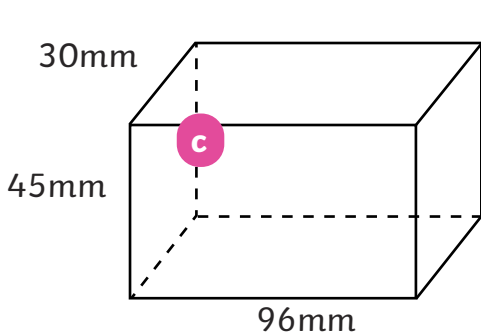
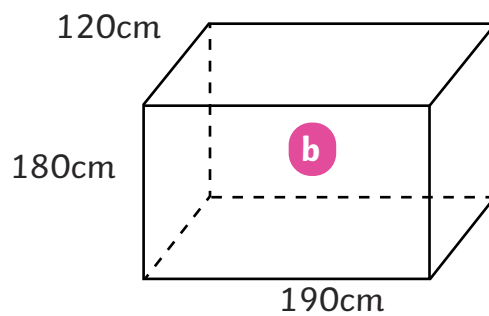
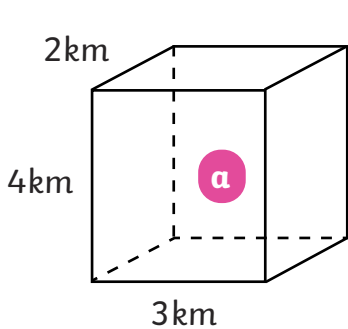
a.

Tick the shape that has the largest volume:



b.

Here are 4 cuboids:



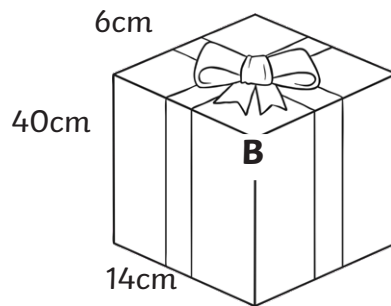
Order the cuboids by volume from largest to smallest

A	B	D	C
---	---	---	---

Largest

Smallest

Here are two gift boxes. Shola wants to work out which box has the greater volume.



not drawn to scale

c. $A = 3000\text{cm}^3$, $B = 3360\text{cm}^3$

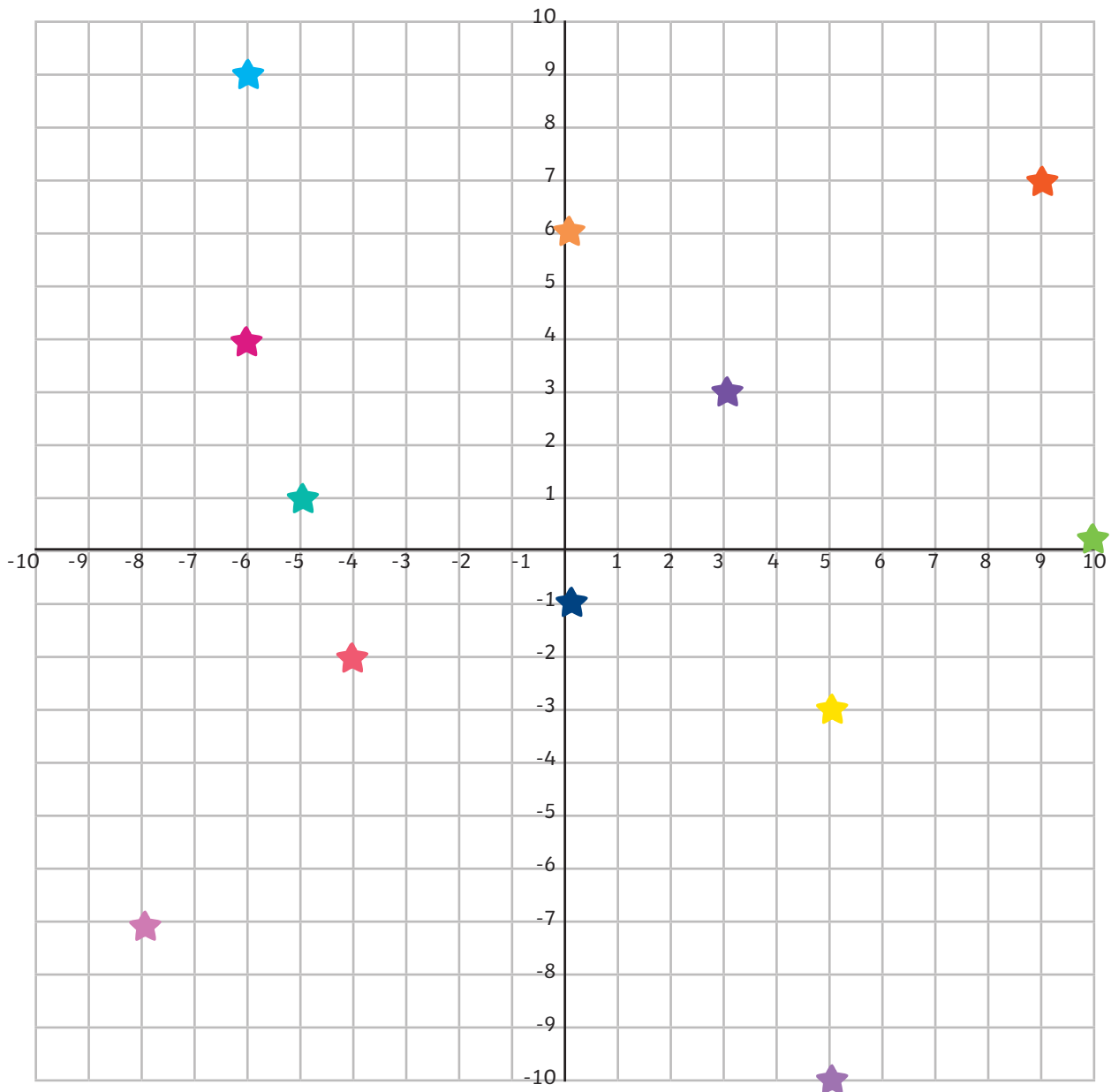
B has the greater volume

Coordinates - Answers

a.

Reading Coordinates

Look at the coloured stars on the grid. Write the coordinates. Don't forget (x, y).



b.

Plotting Coordinates

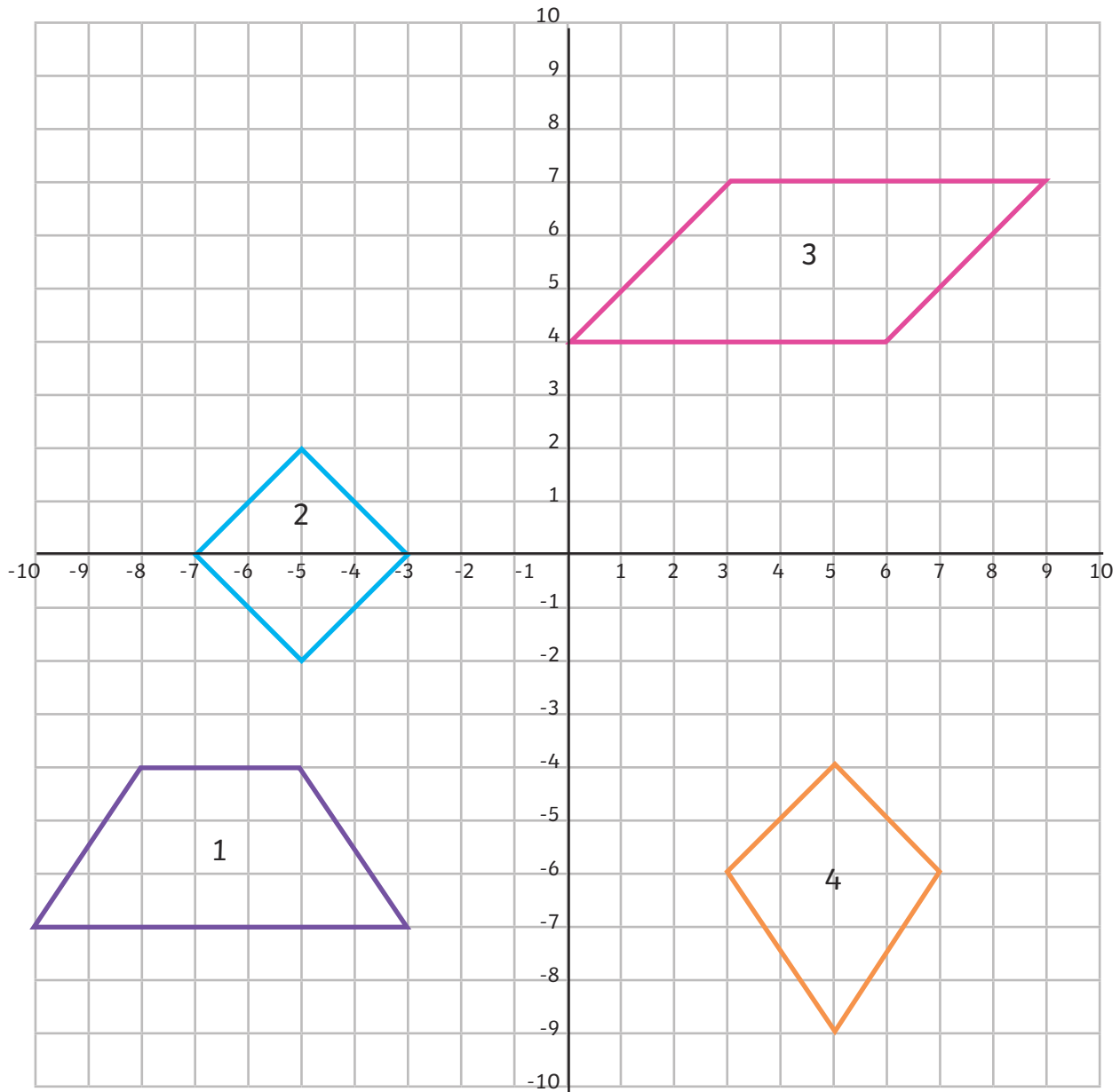
Plot the following sets of coordinates on the grid using a dot or small cross. Connect the dots in each set with straight lines to make a shape.

1. $(-7, -3)$ $(-9, -6)$ $(-2, -6)$ $(-4, -3)$

2. $(-3, 0)$ $(-5, 2)$ $(-7, 0)$ $(-5, -2)$

3. $(0, 4)$ $(3, 7)$ $(9, 7)$ $(6, 4)$

4. $(5, -4)$ $(3, -6)$ $(5, -9)$ $(7, -6)$ $(5, -4)$



c. Shape 1- trapezium

Shape 2- square

Shape 3- parallelogram

Shape 4- kite

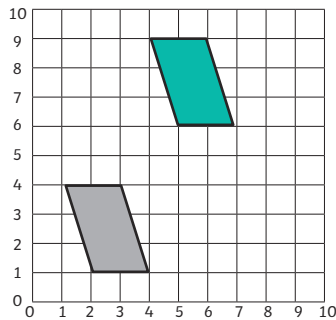
Translations and Reflections - Answers

a. Translating Shapes

1.

This parallelogram is translated **left 3, down 5**.

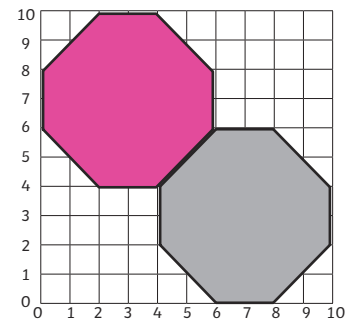
Draw the parallelogram in its new position.



2.

This octagon is translated **right 4, down 4**.

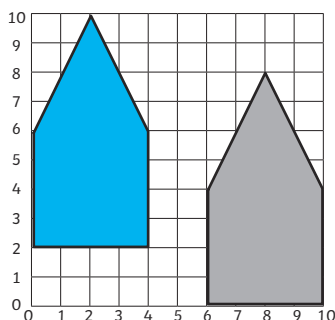
Draw the octagon in its new position.



3.

This pentagon is translated **right 6, down 2**.

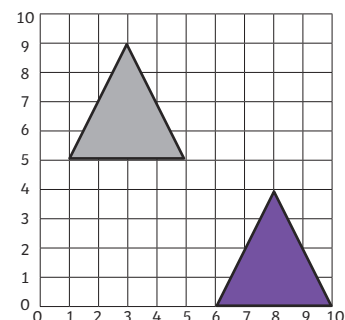
Draw the pentagon in its new position.



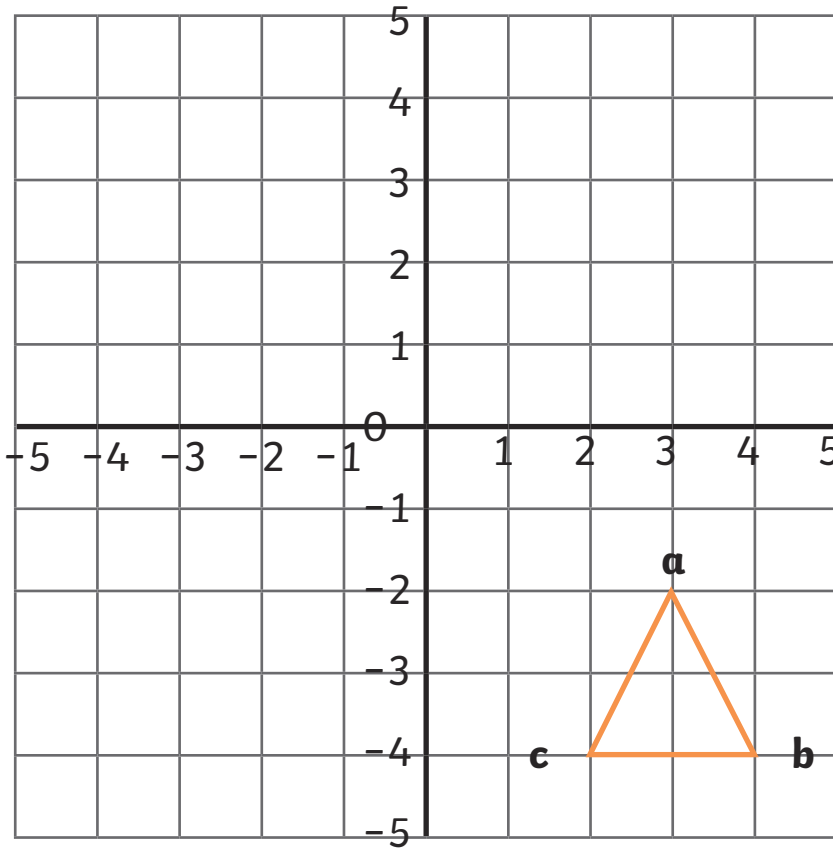
4.

This equilateral triangle is translated **left 5, up 5**.

Draw the triangle in its new position.



b. Translation and Coordinates

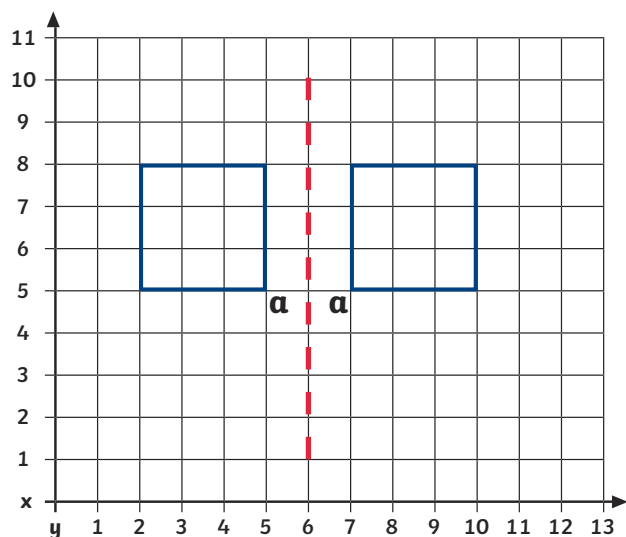


1. If the triangle is translated 5 square left, what will the new coordinates of vertex A be? $(-2, -2)$
2. If the triangle is translated 3 squares up, what will the new coordinates of vertex B be? $(2, -1)$
3. If the triangle is translated 4 squares left and 6 squares up, what will the new coordinates be for vertex C? $(0, 2)$

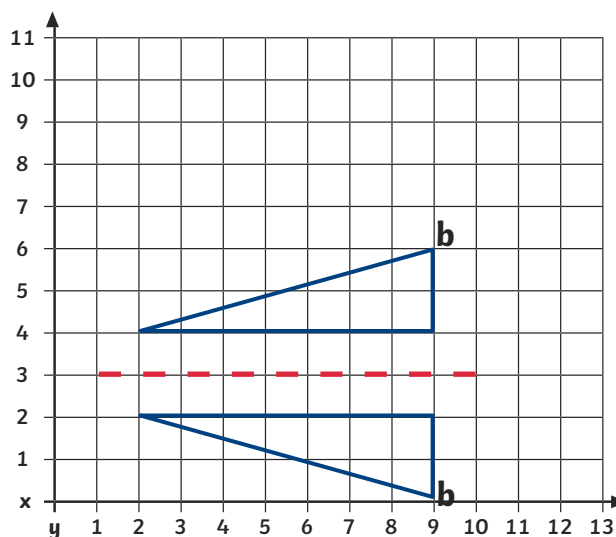
c. Reflecting Shapes

Use the mirror line to reflect the shape and give the new coordinate.

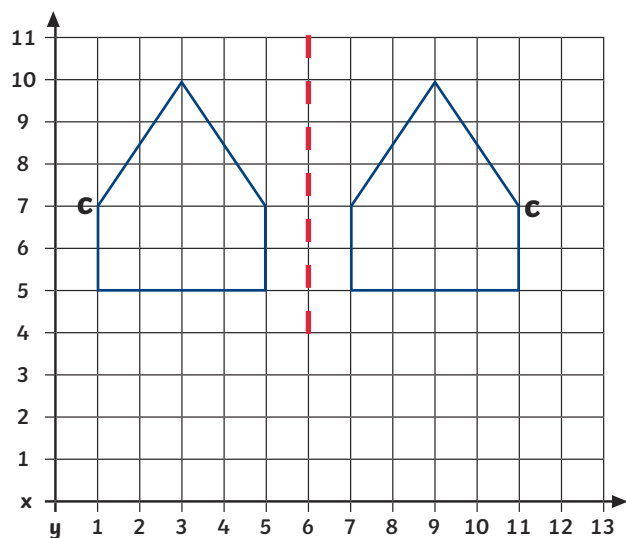
1) New coordinates of corner a = (7,5)



2) New coordinates of corner b = (9,0)



3) New coordinates of corner c = (11,7)



4) New coordinates of corner d = (7,0)

